# The Hitchhiker's Guide to the Treasury Market

## An in-depth overview of the US Government securities market



**Fixed Income Strategy**

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- The US Treasury market remains the largest and deepest fixed income market in the world: as of June 2026 outstanding marketable debt totaled $31.1tn, representing more than a third of the US fixed income market
- We provide an in-depth look at the Treasury market, focusing on the auction process, the types of securities issued by Treasury, the drivers of Treasury yields, as well as relative-value trading strategies in the Treasury market. We also discuss recent developments in market structure and liquidity
- The repo market plays a pivotal role in maintaining liquidity in the US fixed income markets, particularly Treasuries. We provide an overview of the market, discuss recent developments, centrally cleared repo, and explain specialness and fails
- Treasury's debt management strategy aims to finance the federal government at the lowest cost to the taxpayer over time. To that end, one element of Treasury's financing strategy is to be a "regular and predictable" borrower, maintaining relatively stable auction sizes and bringing new issues to market with a set frequency. We review the supply announcement process, how to value when-issued securities, and auction mechanics
- We review Treasury bills, FRNs, TIPS, and STRIPS, discuss who trades these varied Treasury products, and explore their idiosyncratic drivers
- Fair-value frameworks are important in discerning value across the curve and projecting yields over various time horizons. The stance of monetary policy has been the primary driver of Treasury yields over the longer term, and we have found that the relative size of the Fed balance sheet is an important factor for our 10-year fair value model
- We discuss Treasury relative value—the process of determining richness or cheapness along the curve, net of fundamental drivers. We use the level of rates and the slope of the curve to identify macro relative value, and use our par-fitted curve to discern value in individual Treasury securities
- Recent developments in Treasury market structure and liquidity are reviewed. Market depth has improved significantly from the depths associated with the 2022-23 Fed hiking cycle, aided in part by declining volatility, increased trading activity, an ample reserve environment, and support for intermediation stemming from bank deregulation and Treasury's buyback program

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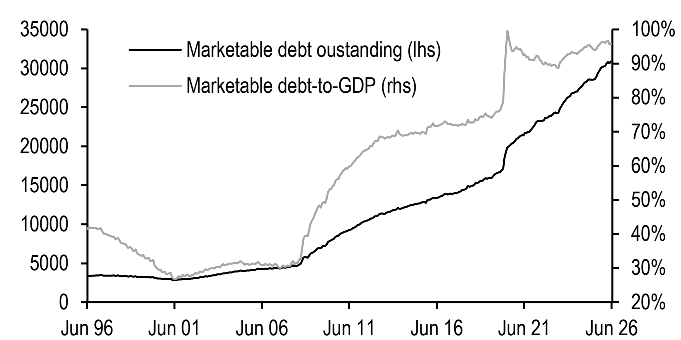
# Introduction

"A national debt, if it is not excessive, will be to us a national blessing," Alexander Hamilton, 1789

The US Treasury market remains the largest and deepest fixed income market in the world: as of June 2026 outstanding marketable debt totaled $31.1tn, representing more than a third of the US fixed income market. The Treasury Department auctions marketable debt in order to fund federal deficit spending, and the market uses Treasuries as a benchmark for pricing and hedging spread asset classes, given the large size, deep liquidity, and risk-free nature of this market. Since the Great Financial Crisis (GFC), the Treasury market has also played a prominent role in the Federal Reserve's implementation of monetary policy: through large-scale asset purchases (LSAPs, also known as quantitative easing), the Fed's Treasury holdings increased more than ten-fold from the onset of the GFC until the end of the COVID-era purchases. The stock of outstanding Treasury debt has grown by over 2.3 times over the past decade. Sharp increases in federal budget deficits during the Great Recession and the COVID-19 pandemic have driven Treasury's increased borrowing needs, which have remained elevated even outside of these cyclical downturns as budget deficits have persisted in recent years (**Figure 1**). Indeed, as deficits have become more structural, the growth in debt has outpaced domestic output, pushing marketable debt-to-GDP close to 100%, from 30% two decades ago.

**Figure 1: The Treasury market has more than doubled in size over the past decade**

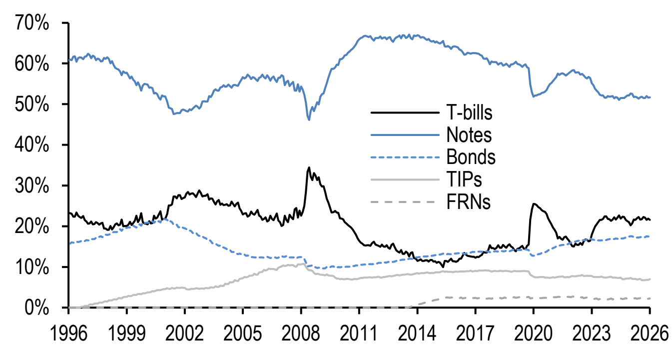
Total outstanding marketable Treasury debt (lhs; $bn) and as a share of US GDP (rhs; %)




Source: US Treasury

**Figure 2: The T-bill share of market debt is at level rarely seen outside of recessionary periods over the last two decades**

Treasury products as share of total outstanding marketable Treasury debt; %




Source: US Treasury

**The Treasury Department auctions five types of marketable securities:** bills, notes, bonds, inflation-protected securities (TIPS), and Floating Rate Notes (FRNs). **Treasury bills** are securities with maturities ranging from a few days to 52 weeks, sold at a discount to their face value. **Notes** are securities that are issued with maturities of 2, 3, 5, 7, and 10 years and pay interest semiannually. **Bonds** mature in 20 or 30 years and also pay interest semiannually. **TIPS** provide investors with inflation protection, as principal is adjusted based on changes in the Consumer Price Index, pay interest semiannually, and are issued with maturities of 5, 10, and 30 years. **FRNs** pay interest that fluctuates based on discount rates for 13-week Treasury bills, are issued for a term of 2 years, and pay interest quarterly. We explore each product in more detail later in this publication.

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**Figure 2** shows that Treasury primarily funds itself using notes, though this share has been declining from its post-GFC peak as Treasury has diversified its suite of products. Notes offer a middle ground between higher rollover risk associated with shorter-dated debt and higher costs associated with longer-dated debt, while its fixed-rate bullet structure is appealing to a wide set of investors. Meanwhile, Treasury typically turns to T-bills for funding during downturns, with the T-bill share of the Treasury market rising to 34% and 25% during the Great Recession and COVID-19 pandemic, respectively. FRNs have accounted for about 2% of the total Treasury market since shortly after the product's inception in 2014. Meanwhile, the TIPS share of the Treasury market has declined over the past two decades, falling from 10% in 2010 to under 7% currently.

**Treasury's primary debt management tenets are to be a regular and predictable borrower and to finance the federal government at the least expected cost over time.** Treasury is transparent to the public in its debt management strategy and utilizes the quarterly refunding process as a way of communicating with market participants. The quarterly refunding announcement is traditionally made on the first Wednesday of February, May, August, and November (or occasionally the last Wednesday of the prior month), and the mid-quarter refunding auctions are held the following week. However, the process begins weeks prior, when the Treasury surveys the primary dealer community on pertinent topics in the Treasury market. Treasury also meets with the Treasury Borrowing Advisory Committee (TBAC) in the days leading up to the refunding announcement. The TBAC is comprised of senior representatives from dealers, asset managers, banks, and hedge funds, and presents recommendations on debt management issues to Treasury.

**Figure 3: The WAM of Treasury's debt has shortened somewhat from its peak earlier this decade, but remains near the longest levels in modern debt management history**

Weighted Average Maturity of US Treasury debt; months


Source: US Treasury

While Treasury aims to be regular and predictable in its borrowings, circumstances sometimes require that Treasury act aggressively to meet its funding needs. As Congress passed a number of stimulus measures to deal with the fallout of the COVID-19 pandemic, Treasury borrowing needs grew significantly and abruptly. Given the magnitude of this borrowing, Treasury could not rely on its suite of coupon-bearing instruments to finance rising deficits in short order, turning instead to T-bill issuance. This in turn shortened the weighted average maturity (WAM) of Treasury's debt sharply. Treasury proceeded to take steps to return its WAM back to pre-pandemic levels, which TBAC had found reduces rollover risk and minimizes the variability of its interest expense, through increasing longer-end auction sizes as deficit needs continued through the post-pandemic recovery. Currently, Treasury's WAM sits at 70.6 months, near the

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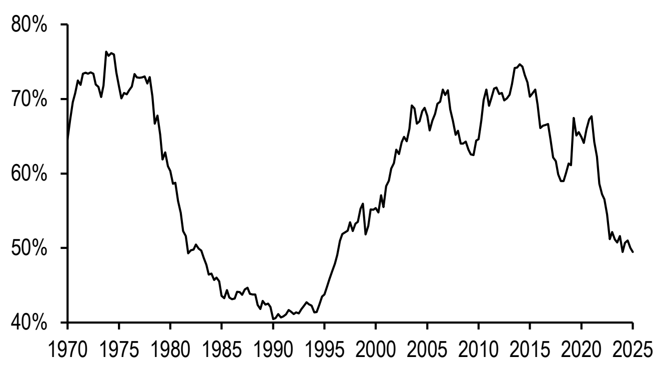
upper end of its historic range (**Figure 3**).

**Ownership of the Treasury market has undergone a seismic shift over the past decade away from more price-insensitive investors.** The Fed, foreign investors, and US commercial banks comprised an average 66% share of the Treasury market for the first two decades of this century, but this share of more price-insensitive investors has plummeted in the post-COVID era, declining to 49% as of 1Q26 (**Figure 4**). The decline in foreign demand has largely been a function of foreign exchange reserves; after rising sharply in the decade through 2013, reserves have been range-bound for the last decade, and the dollar share of those reserves have been falling as well, resulting in a declining official share over the last decade.

The Fed purchased over $3.3tn of Treasury bonds throughout 2020-21 in an effort to stimulate the economy and smooth market functioning through its quantitative easing program, before unwinding this accommodative stance through quantitative tightening, whereby it allowed its Treasury holdings to mature and passively shrink its balance sheet. Through reserve management purchases, the Fed is once more incrementally growing its balance sheet to maintain an ample reserve environment, and in conjunction with re-investments of paydowns from its MBS holdings, is focusing purchases in T-bills. Finally, commercial bank demand was supported by strong deposit growth during the pandemic, but has been declining in recent years as strong loan growth and deregulation has allowed capital to be put to work in higher yielding assets. Marginal demand has instead come from a diversified set of investors, ranging from money market funds, asset managers, and hedge funds (**Figure 5**). This shift away from relatively price-insensitive and buy-and-hold investors alongside increased duration supply in recent years has likely contributed to rising term premium, steeper curves, and increased trading volumes.

**Figure 4: Treasury market ownership has shifted away from more price-insensitive investors over the past decade...**

SOMA, US commercial bank, and foreign combined share of marketable Treasury debt outstanding; %




* Through 1Q26
Source: Federal Reserve Z.1

**Figure 5: ...with an increasing share held by money market funds, asset managers, and hedge funds**

Ownership of Treasury securities (excluding savings bonds) by investor type; %

| | 2010 | 2015 | 2020 | 2025 |
|---|---|---|---|---|
| Foreign Investors | 51% | 47% | 33% | 33% |
| Federal Reserve | 12% | 20% | 24% | 14% |
| Household | 8% | 5% | 6% | 11% |
| State and local govt | 5% | 4% | 4% | 5% |
| Money managers | 5% | 6% | 6% | 6% |
| MMFs | 4% | 4% | 11% | 12% |
| Pension funds | 4% | 4% | 4% | 4% |
| Banking institutions | 3% | 4% | 5% | 7% |
| Insurance companies | 3% | 2% | 2% | 2% |
| Others* | 2% | 2% | 2% | 2% |
| ETFs | 1% | 1% | 1% | 2% |
| Broker dealers | 1% | 1% | 1% | 2% |
| Corporate | 1% | 0% | 0% | 0% |

*Through 4Q25. Other includes GSEs, issuers of ABS, and holding companies
Source: Federal Reserve Z.1

Over the following pages, we provide an in-depth look at the Treasury market, focusing on the Treasury repo market, the auction process, the types of securities issued by Treasury, the drivers of yields, as well as relative value trading. Lastly, we discuss recent developments in market structure and liquidity.

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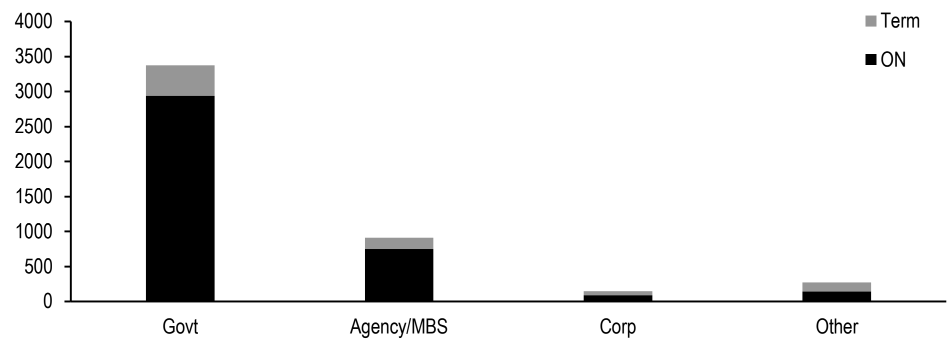
# Repurchase agreements

A repurchase agreement, or repo, is essentially a collateralized loan in which one counterparty posts a financial asset as collateral to borrow cash from a cash-lending counterparty for a short period. At the end of the transaction, the borrower returns the cash to the lender plus interest and receives back the collateral posted at the beginning of the transaction. The repo market plays a pivotal role in maintaining liquidity in the US fixed income markets, particularly the Treasury market. It allows dealers to finance long positions and source securities for setting shorts in order to perform their function as market-maker. In doing so, the dealers intermediate end-user flows, sourcing funds from cash-rich investors to finance their own positions or to on-lend to their clients. Currently, repo secured by Treasury collateral amounts to $3.4tn in securities lent out by primary dealers and $3.0tn in securities taken in by primary dealers, and comprises over 78% of the total repo market (**Figure 6**). Treasury repos usually have very short maturities—nearly 78% mature overnight.

**Figure 6: Over 75% of all repo transactions are collateralized by US Treasury securities**

Repo market collateral and maturity breakdown (securities lent out); $bn




Source: Federal Reserve Bank of New York, J.P. Morgan

Repos allow cash borrowers to leverage their portfolios in order to obtain higher returns and lenders to invest excess cash into a relatively low-risk asset. The primary cash investors in the repo market are money market funds, securities lenders, asset managers, banks, hedge funds, and other long cash investors. The primary cash borrowers are securities dealers, prime brokerage clients, and hedge funds.

Structurally, the repo market can be broken down into two main segments: the bilateral repo market and the tri-party repo market (**Figure 7**). In the bilateral repo market, transactions are done directly between cash borrowers and lenders, and typically involve very specific collateral. In the tri-party repo market, a designated clearing bank (The Bank of New York Mellon) facilitates the collateral selection, custody and settlement between borrower and lender.

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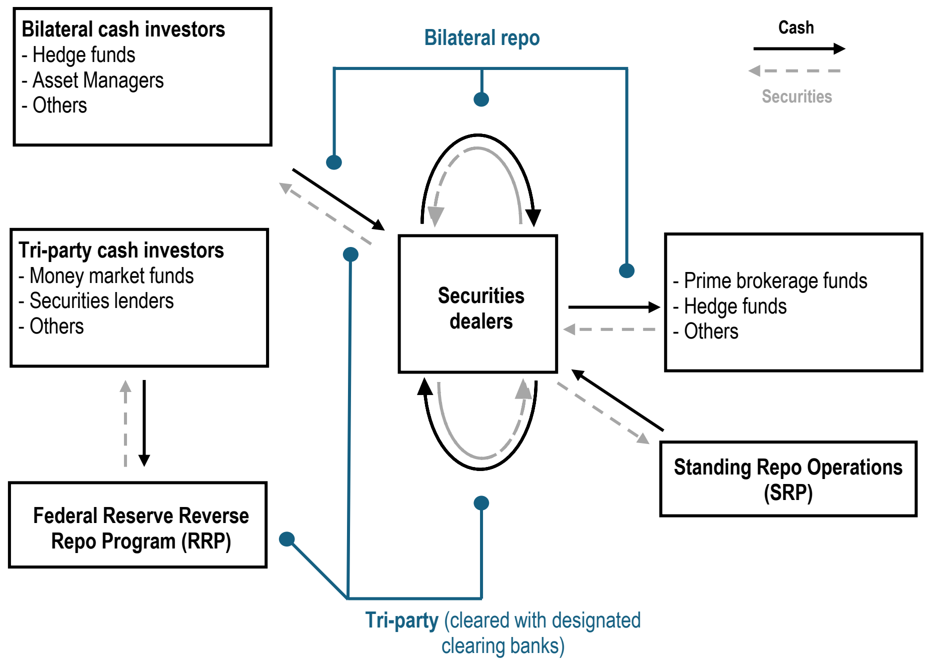
**Figure 7: The repo market is comprised of the bilateral and tri-party repo markets**

Structural diagram of the repo market




Source: Federal Reserve Bank of New York, J.P. Morgan

Repo balances have been trending higher since late 2014, with balances really picking up beginning in late 2018 (**Figure 8**). This has happened for a combination of reasons. First, growth in the budget deficit has led to significantly more Treasury collateral that needs to be financed in the marketplace, at a time when foreign central banks were less involved in buying Treasuries. Second, levered investors were significantly engaged in the cash/futures basis trade, often financing the cash component in the repo markets.

This more recent growth in repo balances has had two important effects: it has increased the importance of centrally cleared repo (see "Centrally Cleared Repo" below) and it has increased the amount of leverage in the system. The latter development proved troublesome in September 2019, when reserves were also declining, leading to a sudden spike in repo rates. It's less of an issue now, as the Fed has been using open market operations to provide additional liquidity via RMPs. Reinvesting MBS paydowns into T-bills effectively removes some Treasury collateral from private investors as well. Separately, the SRP has been useful in providing a soft ceiling to repo rates during periods when repo typically firms, such as settlement dates, month- and quarter-ends, and tax periods.

Treasury repo can trade either as "general collateral" (GC) or "specials." GC repos are repo transactions in which the collateral can vary across a range of specified liquid Treasury securities, while specials are repos secured by a particular Treasury security and trade below GC because of heavy demand for that security. As such, the GC rate is driven by the supply and demand for cash while specials trade based on the supply and demand for specific assets.

The Secured Overnight Financing Rate (SOFR) is an index that provides the broadest market measure of overnight GC repo rates. SOFR is based on a volume-weighted

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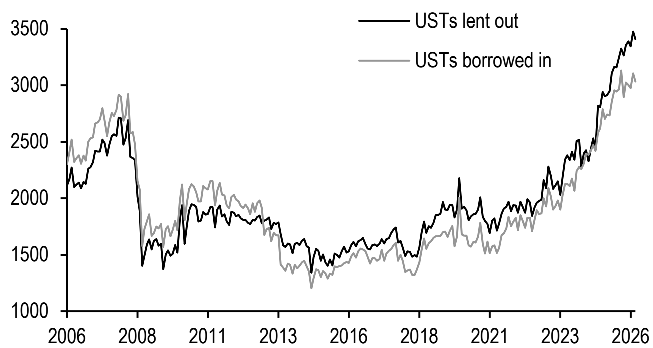
median of overnight GC Treasury repo transactions, including: tri-party repo data collected from the Bank of New York Mellon, inter-dealer general collateral financing data from FICC, and FICC-cleared bilateral repo that drops transactions below the 25th percentile to avoid including "specials" in the calculation (**Figure 9**). Beyond being a measure of repo market borrowing rates, SOFR is also the Fed's preferred reference rate since the LIBOR transition.

**Figure 8: Repo balances have declined since the financial crisis, however balances started to pick back up again in late 2018**

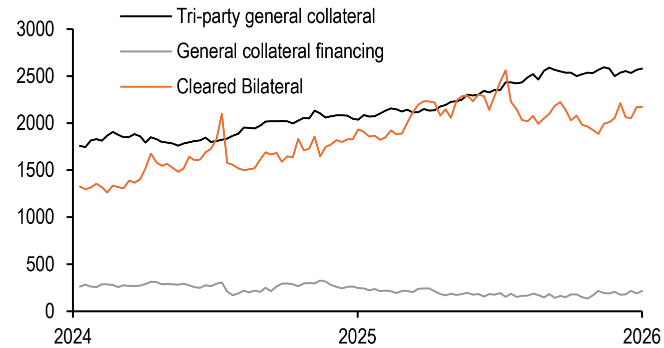
Repo and securities lending outstanding balances*; $bn




Source: Federal Bank of New York, J.P. Morgan
* The Fed did not begin separating repo and securities lending within PD report until 2013

**Figure 9: SOFR is made up of three different sources of repo transactions**

Repo and reverse repo transaction volumes by transaction source; $bn




Source: Federal Reserve Bank of New York, J.P. Morgan

On the specials side of the repo market, dealer hedging and relative-value trading are two factors that drive demand for certain bonds. As such, repo rates for specific issues differ from GC rates in that they are dependent on the supply and demand of a given security. In fact, when specific collateral is in high demand, its repo rate may trade below the GC rate and even negative. When this occurs, the specific issue is considered to be "trading special." The lower a specific issue trades below GC, the more "special" it is.

Several factors influence a given security's "specialness" relative to the GC rate. On the supply side, Treasury plays an important role. Based on its fiscal needs, Treasury can elect to change auction sizes of any Treasury security. Decreases in the stock of a certain security would pressure its repo rate lower and vice versa. Furthermore, regulatory challenges facing dealers can reduce lendable repo supply in some securities as dealers seek to reduce inventories and optimize balance sheet. On the demand side, market sentiment plays a major role. After a large short base is built up around a certain point of the curve, market participants will bid more competitively for specific securities needed to cover their short positions when the market turns.

In a scenario like the above where demand for a specific scarce security is high, it is not uncommon for repo transactions collateralized by the security to "fail" en masse. Consequently, a security's level of specialness and the number of failed contracts in the issue exhibit a strong correlation during periods of immense demand (**Figure 10**). A failed repo occurs when one party of the transaction fails to deliver the collateral specified in the repo's contract in time to settle the trade. There are several negative implications of repo trades failing. Mainly, collateral involved in a failed trade may have already been pledged in a subsequent trade which would cause that trade to fail as well, creating a cascade effect. As a means to prevent fails from occurring and to promote order within the repo market, the FICC charges counterparties who fail to deliver

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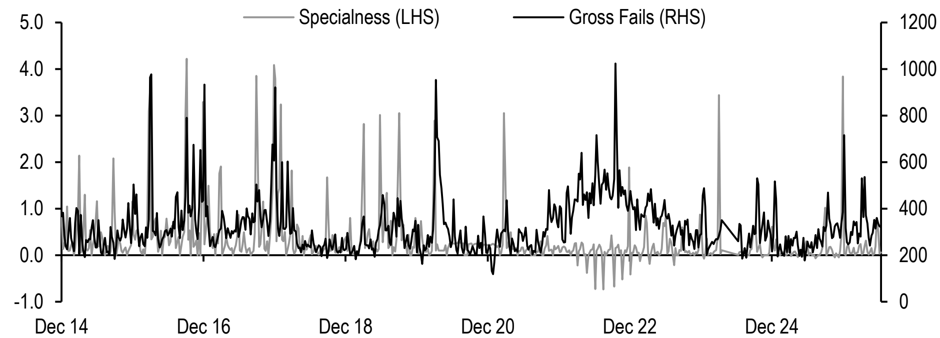
collateral a penalty.

**Figure 10: When demand is high, specialness and failed repo contracts of specific issues exhibit a high correlation**

Specialness of on-the-run 10-year Treasury (%, inverted axis, RHS) vs. gross fails ($bn, RHS)




Source: Federal Reserve, J.P. Morgan

## Centrally Cleared Repo

Sponsored repo is a type of transaction whereby the dealer sponsors a non-dealer counterparty in a cleared bilateral repo trade with FICC (Fixed Income Clearing Corporation). Alongside FICC, the SEC recently approved applications for both CME and ICE to establish covered clearing agencies for Treasury clearing. Functionally, the transaction is similar to a typical delivery-versus-payment (DVP) repo transaction: dealers and end users negotiate on the specific collateral in exchange for cash. However, there are some key operational differences: 1) the trades are novated to FICC, and 2) the sponsoring dealer facilitates all details of the trade with FICC, acting as a processing agent on its client's behalf for all operational functions, including trade submission and settlement with the central clearing counterparty.

There have been recent enhancements to FICC's indirect access framework which have helped address the "double margining" issue for sponsored repo clearing. Most notably, this has been through the collateral-in-lieu model and the Agency Clearing Service—which should further support the transition to Treasury clearing.

One of the benefits of centrally cleared repo for banks is the ability to net those transactions off balance sheet. By engaging in centrally cleared repo, banks can decrease the amount of leverage capital they have to hold. They can also reduce the amount of cross-border repo exposures on their balance sheets (i.e., by sponsoring their offshore clients onto a domestic platform), and accordingly reduce their G-SIB score and potentially the amount of G-SIB capital surcharge they have to hold. In both instances, centrally cleared repo gives banks the ability to free up additional capital on their balance sheets. On the other hand, smaller banks that have to comply with neither G-SIB requirements nor leverage ratio requirements may find limited benefit in engaging in central clearing, especially given the ancillary costs incurred.

For more information, see: *Get in the clear: FAQs on Treasury repo clearing*, 4/25/24 and *Treasury central clearing takes shape via the SEC*, 9/19/22.

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# Supply process

Treasury's debt management strategy aims to finance the federal government at the lowest cost to the taxpayer over time. To that end, one element of Treasury's financing strategy is to be a "regular and predictable" borrower, maintaining relatively stable auction sizes and issuing with a set frequency. The timing of Treasury issuance on an intra-month basis depends on the tenor of the issue being offered:

- **T-bills** are generally auctioned on a weekly basis, with the exception of the 52-week bill, which is auctioned once every four weeks. Thirteen- and 26-week bills are auctioned on Monday; 6- and 52-week bills are auctioned on Tuesday; 17-week bills are auctioned on Wednesday; and 4- and 8-week bills are auctioned on Thursday.
- **Nominal fixed-rate Treasury** auctions are held over three separate weeks. 3-, 10-, and 30-year Treasuries are auctioned mid-month for settlement on the 15th (or the following business day). 2-, 5-, and 7-year auctions are held at the end of the month for settle on the last day of the month (or the following business day). 20-year bonds are auctioned in the week between the two: these bonds carry a mid-month maturity, but settle on either the last day of the month (for newly issued bonds) or the Friday prior to the last day of the month (for reopenings).
- **TIPS** are also auctioned in the week between the mid-month and end-of-month nominal auctions. TIPS auctions settle on the last business day of the month.
- The **2-year FRN** is auctioned at the end of the month. New-issue FRNs settle on the last day of the month, while reopenings settle on the last Friday of the month.

**Figure 11** shows the 2026 auction calendar for longer-term Treasury debt. New-issue 2-, 3-, 5-, and 7-year notes are auctioned monthly, while new 10-, 20-, and 30-year nominals and the 2-year FRN are issued quarterly and subsequently reopened twice. New 5- and 10-year TIPS are auctioned on a semi-annual basis, and new 30-year TIPS are auctioned annually.

**Figure 11: Treasury auctions debt across the maturity spectrum in a "regular and predictable manner"**

Monthly realized and projected gross issuance sizes for coupon Treasuries for 2026, reopenings shaded in grey; $bn

| | 2s | 3s | 5s | 7s | 10s | 20s | 30s | 5y TIPS | 10y TIPS | 30y TIPS | 2y FRN | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Jan 26 | 69 | 58 | 70 | 44 | 39 | 13 | 22 | 0 | 21 | 0 | 30 | 366 |
| Feb 26 | 69 | 58 | 70 | 44 | 42 | 16 | 25 | 0 | 0 | 9 | 28 | 361 |
| Mar 26 | 69 | 58 | 70 | 44 | 39 | 13 | 22 | 0 | 19 | 0 | 28 | 362 |
| Apr 26 | 69 | 58 | 70 | 44 | 39 | 13 | 22 | 26 | 0 | 0 | 30 | 371 |
| May 26 | 69 | 58 | 70 | 44 | 42 | 16 | 25 | 0 | 19 | 0 | 28 | 371 |
| Jun 26 | 69 | 58 | 70 | 44 | 39 | 13 | 22 | 24 | 0 | 0 | 28 | 367 |
| Jul 26 | 69 | 58 | 70 | 44 | 39 | 13 | 22 | 0 | 21 | 0 | 30 | 366 |
| Aug 26 | 69 | 58 | 70 | 44 | 42 | 16 | 25 | 0 | 0 | 8 | 28 | 360 |
| Sep 26 | 69 | 58 | 70 | 44 | 39 | 13 | 22 | 0 | 19 | 0 | 28 | 362 |
| Oct 26 | 69 | 58 | 70 | 44 | 39 | 13 | 22 | 26 | 0 | 0 | 30 | 371 |
| Nov 26 | 69 | 58 | 70 | 44 | 42 | 16 | 25 | 0 | 19 | 0 | 28 | 371 |
| Dec 26 | 69 | 58 | 70 | 44 | 39 | 13 | 22 | 24 | 0 | 0 | 28 | 367 |
| Total | 828 | 696 | 840 | 528 | 480 | 168 | 276 | 100 | 118 | 17 | 344 | 4395 |
| 2025 | 828 | 696 | 840 | 528 | 480 | 168 | 276 | 98 | 115 | 17 | 344 | 4390 |
| Diff. | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2 | 3 | 0 | 0 | 5 |

Source: US Treasury, J.P. Morgan

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## The announcement and when-issue trading

The Treasury issuance process typically begins with the supply announcement. For most Treasury securities, this announcement is made in the week prior to the auction. The auction announcement specifies the size of the offering, CUSIP, issuance and maturity dates, and the type of security to be sold.

After this announcement, the security to be auctioned begins trading on a "when-issued" (WI) basis. Because the coupon on the security is not known during this period, the WI security trades on a yield basis, for settlement on the issue date of the new security. WI trading offers several benefits for the Treasury market. For auction participants, the WI trading period allows for greater price discovery ahead of the actual auction process. For dealers in particular, flows in the WI also allow them to better gauge demand for the upcoming auction. Finally, WI trading helps build liquidity in the new issue.

Because of differences in maturity, coupon, and interest accrual periods, the WIs are valued as a spread to the existing on-the-run issue, known as the WI roll. Buying the WI roll effectively means an investor will sell the current issue for regular settlement to buy the WI for settlement on the issue date. At a high level, roll valuations are driven by four distinct factors (**Figure 12**):

**Figure 12: The WI roll is driven by a combination of carry, curve, liquidity premium, and bad day settlement adjustments**

Fair value projections for WI 3-year Treasury roll; units as indicated

| | |
|---|---|
| Current note | 4.125% Jun-29 |
| Yield of current | 4.18% |
| Expected Amount | $58bn |
| SOMA Add-ons | $6.0bn |
| WI 3-year Maturity | 15-Jul-29 |
| WI Settle Date | 15-Jul-26 |
| Term Repo (%)* | 3.66 |
| Financing Cost (bp) | 0.5 |
| Curve/liquidity (bp) | -0.4 |
| Bad Days (current) | 0 |
| Bad Days (WI) | 1 |
| Bad Day Adjustment (bp) | 0.3 |
| **Estimated Roll (bp)** | **0.375** |
| Expected Range (bp) | 0.125 to 0.625 |

* Term repo through the settlement date
Source: US Treasury, J.P. Morgan

**Figure 13: Example auction bidding structure for a $58bn offering***

Hypothetical example of a Treasury auction process; units as described

| Auction bid (%) | Total size ($bn) | Cumulative bids ($bn) | Filled? |
|---|---|---|---|
| 4.122 | 5.0 | 5.0 | Yes |
| 4.123 | 7.0 | 12.0 | Yes |
| 4.124 | 8.0 | 20.0 | Yes |
| 4.125 | 10.0 | 30.0 | Yes |
| 4.126 | 7.0 | 37.0 | Yes |
| 4.127 | 6.0 | 43.0 | Yes |
| 4.128 | 8.0 | 51.0 | Yes |
| 4.129 | 5.0 | 56.0 | Yes |
| 4.130 | 12.0 | 68.0 | Partial |
| 4.131 | 5.0 | 73.0 | No |
| 4.132 | 7.0 | 80.0 | No |
| Total tendered ($bn) | | | 80.0 |
| Total accepted ($bn) | | | 58.0 |
| Bid-to-cover | | | 1.379 |
| Tenders allotted a high | | | 17% |

* Example assumes no noncompetitive bids
Source: J.P. Morgan

- **Carry/Financing cost:** an investor buying the WI will not accrue interest or pay financing costs until the new issue settles, whereas the investor will give up the carry on the on-the-run over the same period. As a result, the carry for the existing on-the-run for the period between announcement and settlement is added to the WI yield in order to compensate investors for not earning coupon income during the pre-settlement period.
- **Curve:** in upward sloping or inverted yield curve environments, the WI yield will

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differ from the current issue due to the duration difference between the WI security and the existing on-the-run. Changes in the expected coupon of the new issue will also impact the issue's duration and thus curve valuations.

- **Liquidity premium:** on-the-run securities are the most actively traded and liquid Treasury securities, and as a result tend to enjoy a liquidity premium relative to off-the-run securities.
- **Bad day adjustment:** for nominal Treasuries whose maturity date falls on a non-business day, investors will receive the principal and last coupon payment on the next business day (1-3 days after the stated maturity). Because of the delay, bonds with non-business day maturity dates tend to trade with a higher yield. Moreover, this tends to be more impactful for valuations in shorter tenors, as the delay represents a larger fraction of the total life of the security. All else equal, if the new-issue maturity falls on a non-business day, fair value of the roll is more positive, and if the current issue maturity falls on a non-business day, the fair value of the roll is less positive.

## Treasury auction mechanics

On the day of the auction, participants submit bids in either one of two formats: noncompetitive and competitive. Noncompetitive bids are guaranteed to be filled but give bidders no control over pricing—bidders receive their allotment at whatever level the auction clears. Noncompetitive bidders are generally retail in nature, and the share of noncompetitive bids represents a very small portion of the total auction offering.

Competitive bids at auction are conducted in a single-price, or Dutch, auction format. Bidders specify both the quantity to purchase as well as desired yield (to the nearest 0.1bp). The lowest-yielding bids are filled until the auction is completely sold, but all bidders are awarded the highest-yielding bid that is filled. **Figure 13** highlights this process. In the example, aggregate bids at a yield of 4.13% or lower exhaust the $58bn competitive offering size. As such, the auction clearing level is 4.13%, with all competitive bids lower in yield filled in full. At the clearing level, bids are partially filled on a prorated basis such that the total accepted bids equal the size of the offering.

At times, the Fed may also purchase Treasuries at auction. The Fed's System Open Market Account (SOMA) owns more than $4.4tn in Treasuries, and under the existing reinvestment policy, the Fed will reinvest the principal proceeds at auction as these holdings mature. Similar to non-competitive bidders, the SOMA is awarded securities at the same price for all competitive bidders. However, securities sold to the Fed are "added on" to the public offering amount; thus the amount sold to the public remains unchanged, and the additional amount sold to the Fed effectively upsizes the total outstanding of these securities.

For new-issue Treasuries, the coupon is set at the nearest eighth of a percentage point, rounded down, ensuring that the security price is offered at or below par. However, in the event that the auction clears at a yield near zero or at a negative yield, the coupon is floored at 0.125%, and the bond will be issued at a premium. For scheduled reopenings, the spread or coupon is already known at the time of the auction and is thus independent of the reopening auction clearing level. Importantly, as rates fluctuate over time, it is possible that a new-issue security carries the same coupon and maturity date as a pre-existing issue that has rolled down the curve with the passing of time. In this scenario, the auction results in an unscheduled reopening of the older security, instead of trading independently with its own CUSIP.

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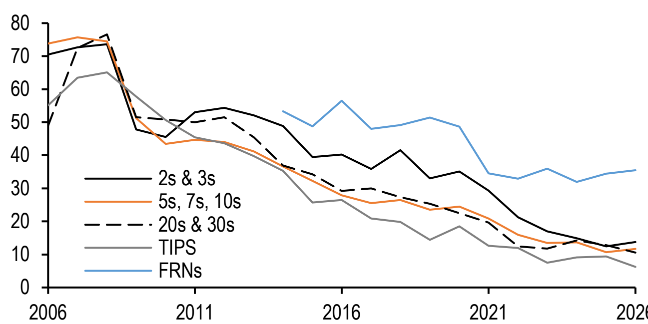
## Who participates at auctions?

Participants at auction fall under two broad categories: primary dealers and end users. Given the sheer amount of debt it must raise on a monthly basis to meet financing needs, Treasury relies on a system of banks and securities broker-dealers, known as primary dealers, in order to help facilitate the auction process. In addition to their responsibilities as trading counterparties for the New York Fed in its open market operations, primary dealers are required to bid for a certain amount of the supply being auctioned. These bids may be placed for the dealers' own books, or on behalf of clients looking to participate in the auction.

Dealer takedown of Treasury supply is also inversely related to the strength of end-user demand, falling as investor appetite for Treasuries rises, and vice versa. Dealer takedown has declined over the past decade, largely driven by greater demand from end users (**Figure 14**). To a lesser extent, dealer participation post-crisis was also adversely impacted by a combination of lower dealer risk appetite and post-crisis regulations, which has made holding Treasury inventory more costly for much of the last decade (see "Market structure and liquidity" below).

**Figure 14: The dealer share of auction allotments has been on a steady decline since the GFC, driven by reduced dealer risk appetite and increased end-user demand...**

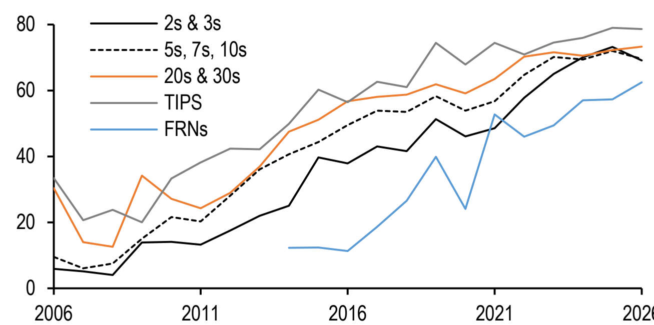
Dealer participation at auction by product type, annual average*; %




*7-year notes introduced in 2009, 30-year TIPS introduced in 2010, and 20-year notes introduced in 2020. 2026 reflects data through June
Source: US Treasury

**Figure 15: ...Investment manager demand has trended higher as assets under management have grown**

Investment manager participation at auction by product type, annual average*; %




*7-year notes introduced in 2009, 30-year TIPS introduced in 2010, and 20-year notes introduced in 2020. 2026 reflects data through June
Source: US Treasury

Investment managers are generally the largest source of demand at auction, particularly in coupon Treasuries (**Figure 15**). Notably, sponsorship from investment managers has trended higher since the GFC, driven by a combination of the growth of Treasuries as a share of the investable fixed income universe, as well as the growth in assets under management. International investors also comprise a large share of Treasury auction allotments. For the official sector, demand is driven primarily by the pace of FX reserve accumulation but can also be influenced by risk appetite. The share of foreign demand at auction declined over the decade prior to the COVID pandemic, and in the years since has remained rangebound at a low share (**Figure 16**). Historically, international investor participation has been lower at long-end nominal and TIPS auctions than in other points along the curve. Treasury's detailed auction allotment data shows that commercial banks, pension and retirement funds, and individuals also purchase Treasuries through the auction process. However, for the most part, these investor classes comprise a very small portion of primary demand.

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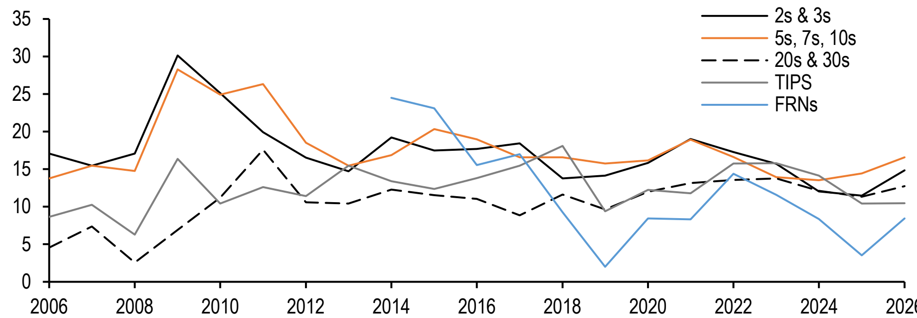
**Figure 16: Foreign participation at auction declined over the pre-pandemic decade, but has stabilized in recent years at low levels**

Foreign investor participation at auction by product type, annual average*; %




*7-year notes introduced in 2009, 30-year TIPS introduced in 2010, and 20-year notes introduced in 2020. 2026 reflects data through June
Source: US Treasury

## Treasury buyback program

Treasury re-introduced its buyback program in May 2024 with two major debt management goals, both tied to reducing borrowing costs over time. First, Treasury established liquidity support buybacks to bolster market liquidity, including by establishing a predictable opportunity for market participants to sell off-the-run securities. Through retiring deep off-the-run securities and replacing with new issue securities, Treasury can help to bolster liquidity in these sectors and reduce warehousing costs for dealers. Second, Treasury established cash management buybacks with the aim of reducing the volatility in its cash balance and bill issuance. Matching redemptions with predictable seasonal fluctuations in deficits can help Treasury to minimize T-bill supply disruptions. In general, buybacks are treated like any other cash outlay for debt management purposes. Importantly, buybacks are not intended to be used to mitigate episodes of acute market stress nor are they to be used to fundamentally change the overall maturity profile of debt outstanding.

Liquidity support buybacks are conducted across nine maturity buckets and are focused on nominal and TIPS, whereas cash management buybacks are focused purely on nominal securities at the very front of the curve (1m-2y). Treasury does not intend to buy back bills, FRNs, or STRIPS. Liquidity buyback operation amounts are distributed evenly across each quarter, are apportioned across buckets based off a pre-set schedule, and Treasury aims to conduct at least one operation per bucket per quarter. Meanwhile, the timing and purchase amounts for cash management buybacks follow a more ad hoc schedule clustered around major tax dates (mid-April, mid-June, mid-September, and mid-December).

Currently, Treasury plans on conducting a maximum of $152bn of liquidity support buybacks and $150bn of cash management buybacks in FY26. In assessing whether changes to purchase maximums in a specific sector are warranted, TBAC has developed a stylized buyback score, which is a simple average of one-year z-scores across buyback offer-to-max, dispersion to spline, and off-the-run discounts in each maturity bucket. A high offer-to-max score for a given operation indicates unmet demand to sell into recent operations. Elevated dispersion to a fitted Treasury spline curve (i.e. RMSE) suggests off-the-runs are not trading efficiently. Finally, a growing discount between off-the-runs and their on-the-run counterparts could indicate a deterioration in market functioning. **Figure 17** recreates this buyback score across maturity buckets, showing that buyback

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operations are likely well sized, while buyback sizes are more than sufficient in the belly of the curve.

**Figure 17: Treasury's buyback scores indicate operations are likely well sized, while buyback sizes are more than sufficient in the belly of the curve**

Stylized buyback scores for Treasury buyback sectors

| | | Buyback offer to max | | RMSE (bp)* | | Off-the-run discount (bp)** | | |
|---|---|---|---|---|---|---|---|---|
| | Tenor | Current | 1y z-score | Current | 1y z-score | Current | 1y z-score | Buyback score |
| Nominal coupons | 1m-2y | 6.9 | 0.9 | 1.3 | -1.2 | -0.63 | 0.4 | 0.1 |
| | 2y-3y | 2.2 | 3.1 | 0.7 | 0.1 | -0.42 | -1.2 | 0.7 |
| | 3y-5y | 2.4 | -1.4 | 0.9 | 0.0 | 0.28 | -2.0 | -1.1 |
| | 5y-7y | 1.3 | -1.6 | 0.4 | -1.2 | -0.48 | -1.8 | -1.5 |
| | 7y-10y | 1.1 | -0.3 | 1.7 | 2.2 | -0.66 | 0.7 | 0.9 |
| | 10y-20y | 9.0 | -1.5 | 1.1 | -0.3 | -0.62 | -0.2 | -0.7 |
| | 20y-30y | 14.3 | -0.7 | 0.6 | -1.1 | 0.56 | 0.9 | -0.3 |
| TIPS‡ | 1y-10y | 4.1 | 0.2 | 2.2 | -0.4 | -1.03 | 1.5 | 0.5 |
| | 10y-30y | 1.5 | -1.2 | 2.6 | -1.5 | -2.32 | -1.5 | -1.4 |

* RMSE for nominal Treasury and TIPS spline fitted curves, by sector
** For nominal Treasuries, defined as 4x-old asset/current swap spread differential for hot run point. For TIPS, defined as 2x-old/current IOTA differential (based on z-spreads of TIPS and the nominal comparator)
† For TIPS, positions in 0-6Y are shown for shorter-maturity bucket, and positions >6Y are shown for longer-maturity bucket
‡ TIPS buyback buckets adjusted from 1y-7.5y to 1y-10y and 7.5y-30y to 10y-30y in 3Q25. Data prior to 3Q25 refer to the old bucket definitions
Source: Federal Reserve Bank of New York, J.P. Morgan

In an effort to be regular and predictable, Treasury announces the next quarterly buyback schedule as part of the quarterly refunding announcement. Treasury aims to be a price-sensitive buyer, and actual buyback purchase amounts will depend on market conditions, with the possibility that Treasury will not buyback any debt for a given operation. Treasury announces a list of eligible CUSIPs prior to the operation, subject to certain purchase limits. The free float of a security - which is the amount outstanding less SOMA ownership and stripped amount - must exceed $10bn for nominals and $5bn for TIPS. Moreover, SOMA ownership must not exceed 70% for any security. Furthermore, Treasury excludes securities which are or are close to being on-the-runs/ CTD for futures contracts, securities close to coupon payment dates, Treasuries trading significantly special in repo or at a significantly lower yield than T-bills, and Treasuries with coupons that mature around quarterly tax payment dates. As the fiscal agent of the United States, the New York Fed conducts Treasury buyback operations as multiple price reverse auctions with primary dealers and other eligible counterparties on behalf of Treasury.

# Product overview

## 1. Notes and bonds

As discussed earlier, notes and bonds make up the lion's share of the Treasury market: in the decade prior to the pandemic they consistently accounted for between 70% and 80% of the outstanding marketable debt, but now represent a reduced 68% share, towards the lower end of ranges in recent decades outside of crisis periods. That said, notes and bonds still represent roughly 80% of daily trading volumes in the Treasury market.

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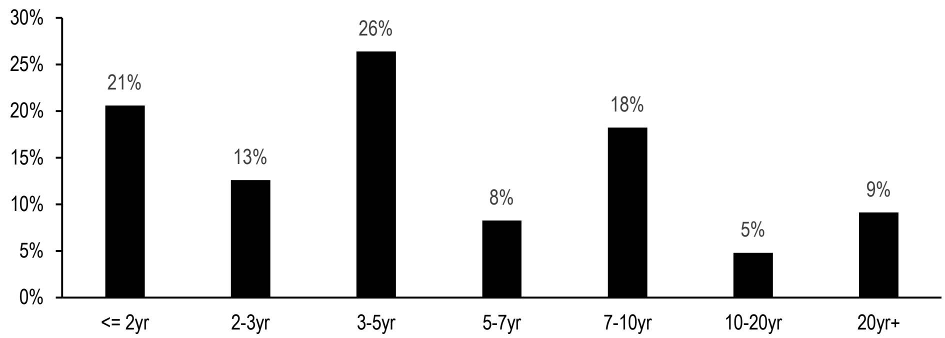
**Figure 18** shows the distribution across various sectors, and illustrates that in notional terms, debt maturing in 3- to 5-years comprises the largest share of daily trading volumes across the curve, led by the under 2-year and 7- to 10-year sectors. This makes sense, as the duration of many investment grade bond indices are close to 6 years and has been declining in recent years. While notes and bonds represent the majority of trading and stock in the Treasury market, we detail all other products here, and then discuss our valuation framework for nominal Treasuries below (see "Key drivers of yields" below).

**Figure 18: Treasury notes and bonds account for more than 80% of trading in Treasury securities, with the bulk focused in intermediates**

Average daily trading volumes in Treasury coupon securities by sector as a share of total Treasury trading volumes, 2026 average*; %




*Through 6/30/2026
Source: TRACE

## 2. Treasury bills

Unlike other Treasuries, T-bills are not coupon-bearing instruments: they are sold at a discount to their par amount and pay interest at maturity. The interest paid is equivalent to the par value minus the purchase price of the T-bill. The Treasury Department auctions T-bills in seven main maturities all under 1-year in term: 4-week, 6-week, 8-week, 13-week, 17-week, 26-week, and 52-week. Thirteen-, 26-, and 52-week bills are announced on Thursday for auction the following week, while 4- and 8-week bills are announced on Tuesday for auction on Thursday. Seventeen week bills are also announced Tuesday, for auction on Wednesday (**Figure 19**).

In addition, Treasury also auctions Cash Management Bills (CMB) in various tenors on an ad hoc basis, in order to help manage its cash flows when the traditional suite of bills does not adequately meet these needs. As its financing needs have grown and the T-bill share has increased, Treasury has innovated through the expansion of its suite of T-bill offerings, introducing the 8-, 17-, and 6-week bills over the past decade.

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**Figure 19: Most T-bills are auctioned weekly, though the 52-week bills are auctioned every four weeks**

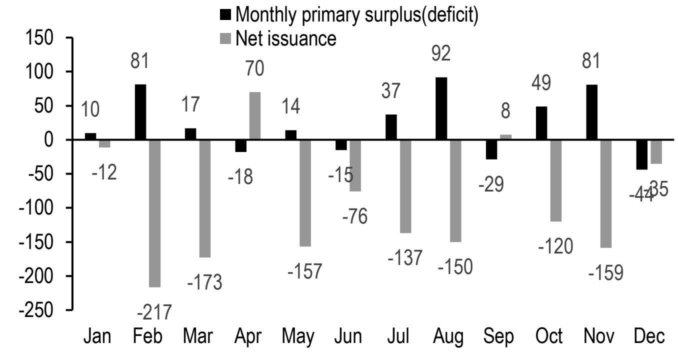
Treasury bills auction schedule

| Term | Aucion Frequency | Announcement | Auction |
|---|---|---|---|
| 4-week | Weekly | Tuesday | Thursday |
| 6-week | Weekly | Thursday | Tuesday |
| 8-week | Weekly | Tuesday | Thursday |
| 13-week | Weekly | Thursday | Monday |
| 17-week | Weekly | Tuesday | Wednesday |
| 26-week | Weekly | Thursday | Monday |
| 52-week | Every four weeks | Thursday | Tuesday |

Source: US Treasury

**Figure 20: Seasonally, the federal government tends to run larger deficits in months where tax receipts are low and T-bills fill the net gaps during these periods**

Monthly primary surplus/(deficit)* and monthly Treasury bill net issuance; average of FY05-FY25; $bn




*Primary surplus/(deficit) is revenues less expenditures, excluding interest payments
Source: US Treasury

Treasury primarily uses T-bills as a cash management tool. Seasonally, the smallest primary deficits come in months when Treasury receives corporate tax receipts (March, June, September, and December), estimated individual tax payments (June, September, and January), and individual income tax receipts (April), while primary deficits are larger in all other months. Conversely, T-bill issuance tends to mirror this pattern, with net issuance highest in months in which the federal government runs the largest primary deficits (**Figure 20**). What creates this need? Treasury's issuance calendar in notes, bonds, TIPS, and FRNs varies only slightly from month to month, in an effort to remain a reliable and predictable borrower, and it is easier to increase or decrease bill issuance on a weekly or monthly basis.

**Figure 21: The foreign share of T-bill ownership declined through the 2010s alongside a rise in MMF sponsorship, and the Fed's T-bill share remains small compared to its pre-GFC share**

Treasury bill ownership by investor type; %

| Investor Type | 2000 | 2005 | 2010 | 2015 | 2020 | 2025 |
|---|---|---|---|---|---|---|
| Federal Reserve | 28% | 28% | 1% | 0% | 7% | 4% |
| Insurance Co. | 1% | 5% | 2% | 2% | 1% | 3% |
| MMFs | 9% | 7% | 16% | 16% | 42% | 41% |
| Mutual Funds | 0% | 0% | 3% | 3% | 1% | 0% |
| Foreign | 27% | 28% | 40% | 48% | 21% | 22% |
| Other | 35% | 32% | 39% | 31% | 28% | 30% |
| Total | 100% | 100% | 100% | 100% | 100% | 100% |

Source: Federal Reserve

Who trades Treasury bills? **Figure 21** shows that money market funds and foreign investors remain the dominant players in the market, together owning just over 60% of the stock of outstanding T-bills. Specifically, a series of money market reforms in the wake of the GFC spurred a large rotation out of prime funds into government funds, increasing the MMF presence in the T-bill market. Meanwhile, the Fed owns just 4% of the market, well below the 28% share which stood prior to the GFC. During the throes of the financial crisis, the Fed ran down its short-term holdings in order to ramp up its

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liquidity facilities, and then only focused on buying longer-term Treasuries during its large-scale asset purchase programs. As discussed above, the Fed's ownership of T-bills has been rising via RMPs and reinvesting MBS paydowns into T-bills.

## 3. Treasury FRNs

Two-year FRNs are daily-reset floaters, indexed to the High Rate from the most recent 13-week Treasury bill auction (applied with a 1-day lag), and interest is paid quarterly. Each day's interest accrual is calculated based on the reference rate plus a spread and is subject to a 0% floor. The 13-week bill auction occurs weekly on Mondays, or the first business day of the week, so although the accrual rate resets each business day, the reference rate changes only once per week. The spread is determined in the initial FRN auction and is the number of basis points necessary to set the initial auction price close to par. It can be positive, zero, or negative and is rounded to the nearest tenth of a basis point. Once determined, the spread will not change over the life of the security—the spread on reopened FRNs will be the spread determined in the first auction of that security.

Additionally, there is a 2-day lockout period prior to the auction settlement date and quarterly interest dates: a 13-week bill auction that becomes effective during this period is ignored until after the issue date or interest date, and the rate in effect will instead reference the previous 13-week bill auction.

Treasury auctions new FRNs quarterly, with two monthly reopenings in subsequent months—currently in the sizes of $30bn and $28bn, respectively. Original issue offerings settle on the last calendar day of a month (or the first business day thereafter), while reopenings settle on the last Friday of a month (or the first business day thereafter). The auction process and mechanics are similar to other government securities, with Treasury announcing the offering on the Thursday prior to the auction, allowing for a WI trading period, and FRNs are auctioned in a single price format (see Treasury auction process: mechanics and participants). In contrast though, competitive bids are submitted in terms of a desired discount margin relative to the reference rate, rather than yield.

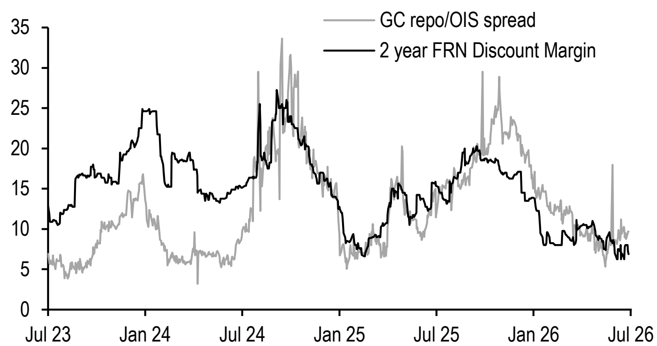
**Relative pricing**

In thinking about relative pricing of Treasury FRNs, we can consider recreating the same cash flow stream: an investor could buy 13-week bills and roll them every week for the next two years, thereby earning the same as the FRN index rate. The extra cost of rolling over T-bills involves paying bid-offer to stay current each week, suggesting that the FRN DM could theoretically trade in deeply negative territory to make an investor indifferent. That said, through rolling T-bills rather than locking in the same index for term through a FRN, the investor has retained the opportunity to redeploy capital. To compensate the investor for pre-committing lending for two years, the FRN DM is often positive, and can be thought of as a form of term funding premium (see *The one with swap spreads*, Ipek Ozil, 1/8/26).

Theoretically, we would expect that Treasury FRN DMs tighten as the OIS curve steepens; as markets expect a higher Fed funds rate in the future - and thus a higher T-bill reference index rate - the spread on FRNs tighten, reflecting increased demand for shorter-duration assets. To an extent this is true, but we also find other technical factors are also significant drivers of FRN spreads. **Figure 22** shows that DMs on 2-year FRNs tend to be directional with term GC repo/OIS spreads, widening as balance sheet costs rise, and vice versa. This makes intuitive sense: Treasury FRNs trade less frequently than T-bills, and are a more inventory-driven product. Thus, as repo widens to OIS and balance sheet becomes more expensive, DMs tend to widen out. We also find that FRN DMs tighten when Government MMF WALs extend; Government MMFs own 72% of the Treasury FRN market, and considering these floaters carry a 1 day WAM, money market funds primarily face WAL rather than WAM constraints when considering their investments into the product. Together, we have found that 1m/1y1m OIS curve, 3-month GC/OIS spread, and Government MMF WAL explain roughly 83% of the variation in Treasury FRN DMs over the past two years (**Figure 23**).

**Figure 22: Treasury FRN DMs tend to be directional with funding conditions as they are more inventory-driven product...**

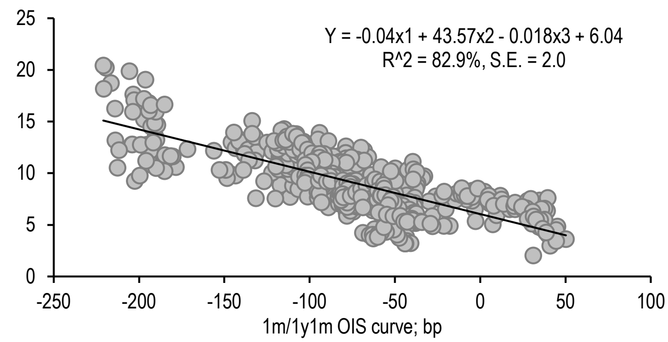
2-year Treasury FRN DM and 3-month GC/OIS spread; bp




Source: J.P. Morgan

**Figure 23: ...which alongside the slope of the money market curve and Government MMF WALs, can explain much of the recent variation in spreads**

2-year Treasury FRN DM regressed on 1m/1y1m OIS curve (bp), 3-month GC repo/OIS spread (bp), and Government MMF WAL (days), regression over the last two years; bp

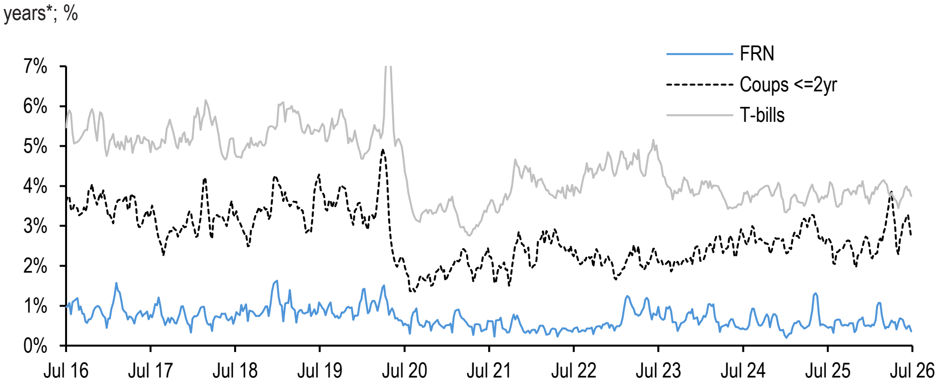



Source: J.P. Morgan, Crane

Relative pricing also reflects differences in the liquidity of each product. Average daily trading in FRNs is roughly 2% of T-bills and roughly 1% of short coupons. Even after adjusting for the diminutive size of the floating rate market, FRN turnover is still dwarfed by similar-maturity T-bills and short coupons, underscoring the buy-and-hold nature of the investor base (**Figure 24**). Therefore, the low relative liquidity of FRNs also likely contributes to the persistent discount relative to bills.

**Figure 24: FRN turnover is significantly lower than that of other short-duration Treasuries**

Average daily trading volume as a fraction of total outstanding for FRN, T-bills, and coupons with maturity less than 2-years*; %




*Coupon trading volumes within the 0- to 2-years bucket whereas outstanding is within the 0- to 1.5-year bucket

Source: Federal Reserve Bank of New York, J.P. Morgan

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## 4. TIPS

**TIPS are structured to provide inflation-protected cash flows to investors.** Thus, an investor that purchases a TIPS outright locks in a real return that is not a function of future inflation. Inflation protection is achieved by indexing the principal of a TIPS to the US headline CPI non-seasonally adjusted series (CPURNSA Index on Bloomberg) published by the Bureau of Labor Statistics (BLS)—the principal of a TIPS increases with inflation and decreases with deflation. Each day, the inflation-adjusted principal is equal to par multiplied by the inflation index ratio, where the index ratio is calculated as the reference CPI on the trade settlement date divided by the reference CPI on the first issue date (the base CPI value).

Since CPI for a given month is released about two weeks after the end of that month, and CPI must be known prior to issuing a TIPS, the **principals are indexed to CPI with a 3-month lag**. Specifically, the reference CPI on the first day of a calendar month is based on the CPI for the third preceding calendar month. For example, the principal on June 1 is based on the CPI reported for March, and the principal on July 1 is based on April CPI. All rounding is to 5 decimal places. On any day between June 1 and July 1, the reference CPI value is calculated by linearly interpolating between the March and April CPI values.

Semiannual coupon payments are calculated as a fixed percentage of the inflation-adjusted principal. **At maturity, holders of TIPS are paid the adjusted principal or the original principal (par), whichever is greater.** Effectively, this means that TIPS have an embedded floor on the principal amount.

TIPS are quoted on a **real price/real yield basis**, and real yields are derived in the same way as fixed-coupon nominal Treasuries, using the "real" cash flows (real price, real coupons, and par at maturity) to calculate a real Internal Rate of Return ("IRR"). When making a cash transaction in TIPS, the real price of the bond (plus the real coupon accrued) is adjusted by the daily index ratio. A list of daily TIPS index ratios can be found on *Treasury's website* and is available in our daily *TIPS Fitted Curve Relative Value Report* .

**Carry** is defined as the income accrued while holding a security minus the cost of funding the position. In the case of conventional fixed-coupon bonds, this can be seen in simple terms as the coupon accrued minus the cost of funding the bond investment (i.e. its repo rate). The difference for inflation-linked bonds is that the dirty price of a linker includes the accrual of a real coupon plus the accrual of inflation for the given horizon. Since the cash flows of inflation-linked bonds are indexed, this means that holding a linker for a given time gives you a positive carry effect during periods when (lagged) inflation indices increase and a negative carry effect when indices decrease. The actual carry is thus the real coupon accrued plus inflation accrual minus the cost of funding. The normal evolution of inflation indices is what makes carry in inflation linked bonds particularly interesting. The seasonal variation in inflation makes carry on inflation-linked bonds very volatile, switching from positive to negative throughout the year.

**What is the TIPS "breakeven"?**

Breakeven inflation can simply be thought of as the average annual inflation needed over the life of the TIPS in order for it to provide the same return as a Treasury with the same maturity (i.e., the level of inflation that makes an investor indifferent between buying TIPS or Treasuries). We approximate this breakeven rate as the similar-maturity

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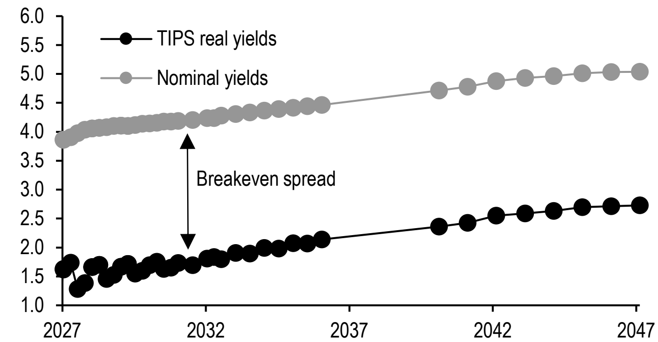
Treasury yield minus TIPS real yield (usually it is quoted in basis points). **Figure 25** illustrates the breakeven curve.

**Figure 25: The breakeven is the spread between a TIPS real yield and a similar-maturity Treasury yield…**

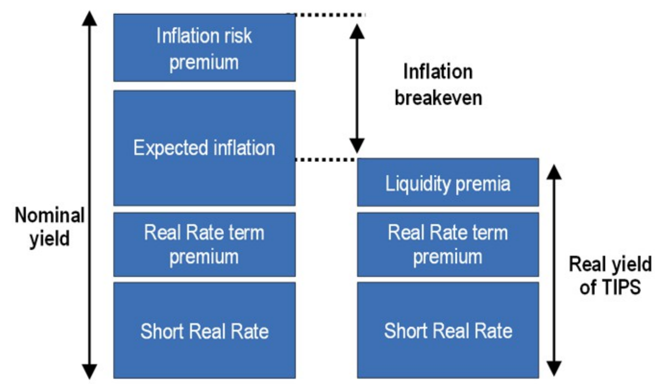
TIPS real yield curve and yield curve of similar-maturity nominal Treasuries; %




Source: J.P.Morgan

**Figure 26: …and this "breakeven" rate reflects more than just expected inflation**

Breakdown of nominal and real TIPS yields




Source: J.P.Morgan

Nominal and real yields can be decomposed into various factors as shown in **Figure 26**. It is worth noting that **although inflation breakevens are a good measure of the market expectations of inflation, and are often used in modeling for such a purpose, there are other market-driven factors that determine breakeven rates**. Specifically, the inflation breakeven can be rewritten as expected inflation plus an inflation risk premium minus a liquidity premium, and each factor can vary across maturity points. Real TIPS yields reflect a liquidity premium (or illiquidity cost) to compensate investors for the relative illiquidity of inflation-linked bonds compared to nominals. As we saw during the 2008 financial crisis and again in March 2020, this component can sometimes become a dominant driver of breakevens.

Still, market participants, as well as the Federal Reserve, use TIPS breakevens as a market-based measure of inflation expectations. In particular, investors and the Federal Reserve regularly monitor the 5Yx5Y forward breakeven measure derived from par Treasury rates. Compared to spot breakevens, the Fed views 5Yx5Y breakevens as less affected by cyclical factors such as energy prices and thus a better measure of how well the market thinks it is meeting its goal of longer-term price stability (see *Dropping anchor: Why five-year forward five-year inflation has stayed rangebound*, 06/09/26 and *Why we look at 5-year forward 5-year inflation*, 3/23/11).

A breakeven position is established by buying (selling) TIPS and selling (buying) a DV01-matched amount of Treasuries. Usually a 1:1 risk ratio is used, but other ratios can be used if the investor believes TIPS will be more/less volatile than Treasuries. For example, investors who want to position for the outperformance of TIPS while limiting their duration exposure, may choose to weight the short nominal Treasury leg of a breakeven widener trade, based on the empirical beta of real yields versus nominal yields. In addition to trading breakevens themselves, investors can also take views on the breakeven curve, or do breakeven switches.

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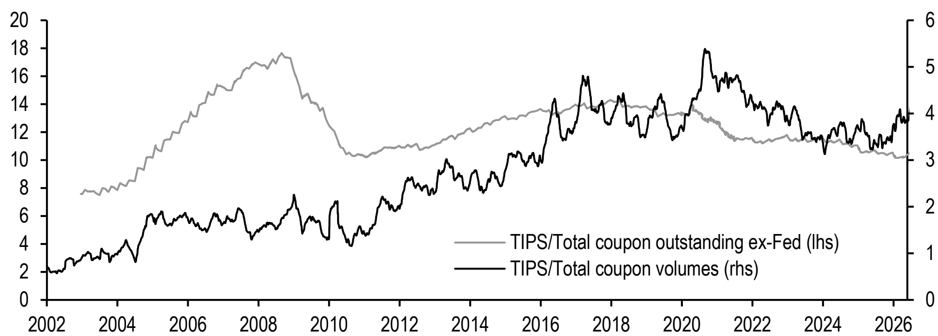
### How liquid are TIPS?

The TIPS market has been around for almost thirty years and comprises a significant share of the Treasury market, TIPS remain far less liquid than their nominal counterparts. Over recent years, daily trading volumes with primary dealers have risen to an average of $24bn/day in 2026, but still equating to about 1% turnover each day. In contrast, daily trading volumes in nominal coupon Treasuries have been around $606bn/day, or about 3% turnover, and TIPS have represented roughly 4% of total coupon volumes (**Figure 27**). The liquidity in TIPS relative to nominals has decreased since 2021 as TIPS issuance has not increased as quickly as it did for nominal notes and bonds. Additionally, a greater share of buy-and-hold investors likely contributes to the lower turnover.

**Figure 27: TIPS are not as liquid as nominal Treasuries**

Average daily TIPS trading volumes as a fraction of total coupon Treasury volumes (LHS) and TIPS outstanding (ex-Fed) as a fraction of total coupon Treasuries outstanding (ex-Fed) (RHS); % both axes




Source: US Treasury, Federal Reserve Bank of New York

### Who trades TIPS?

In the early years of the TIPS program, dealers were the largest buyers of TIPS at auction, but over the last 15 years, end-user demand for TIPS has steadily increased (**Figure 28**). Specifically, investment-manager demand for TIPS has risen since the financial crisis. TIPS are *not* included in commonly-followed benchmark aggregate indices, but some core bond funds using these indices may still choose to make strategic allocations to TIPS. In addition, some investment managers may have dedicated TIPS or inflation-protected funds and may buy (sell) TIPS when these funds see inflows (outflows) in order to match the profile of their benchmark index. As **Figure 29** shows, assets under management for inflation-protected bond funds grew sharply after the inflation shock, peaking post-COVID-19 in December 2021 at $416bn before declining to $300bn in October 2023 and then rising more recently after a wave of global tariffs and the conflict between the US and Iran in 2026. Inflows have been predominantly focused in short-dated funds in recent years, and our work has shown that most inflation-protected funds maintain an underweight TIPS exposure, often choosing to allocate out of benchmark in search of excess returns (see *The inflation fund landscape: reviewing trends in flows, composition, and positioning*, 5/28/24)

Although **foreign investors** buy a smaller portion of TIPS at auction, anecdotal evidence suggests that central banks and sovereign wealth funds are major players in the market. Traditionally, foreign central banks are mostly buy-and-hold investors, and operate with longer investment horizons than other real money investors. They may or may not manage their TIPS holdings against a benchmark index. Foreign holdings of

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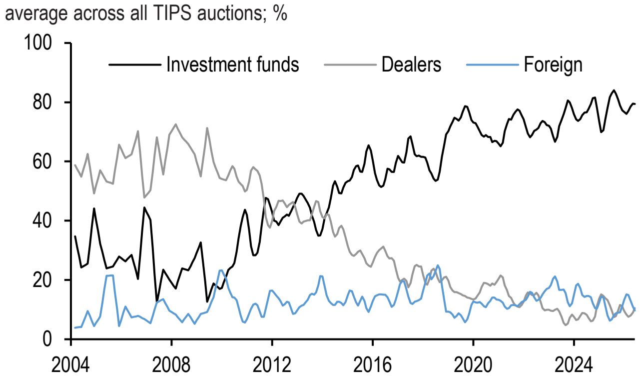
TIPS are available annually from the Treasury International Capital System (TIC), and these data show that foreign investors accounted for 39% of the market as of June 2025, with China representing the largest share. TIPS now comprise roughly 11% of foreign official Treasury portfolios, the highest share on record. This reflects a broader trend in recent years whereby foreign official institutions have increased the proportion of their Treasury holdings dedicated to TIPS, FRNs, and short-term debt at the expense of long-term nominal holdings. Meanwhile, the foreign private and official institutional shares of the TIPS market have converged to roughly 19%, which represents a record high for the private community but represents a material decline for the official community from a peak of 32% in 2015.

**Hedge funds** are also large players in the TIPS market. They may use TIPS to express a view on real yields, the real curve, bond spreads, TIPS breakevens, and the breakeven curve. They may also opportunistically buy/sell TIPS to earn carry. Hedge funds are also very active in the very front end of the curve (<1Y to maturity). Real money accounts benchmarked to TIPS indices tend to sell their front-end TIPS as soon as they roll out of the index, and hedge funds tend to buy this paper when it cheapens.

The **Fed** can be a large driver of demand for TIPS. Given the aggressive pace of Fed purchases throughout QE4, over $482bn TIPS (inflation-adjusted) were held in the SOMA portfolio at its peak in April 2023, representing more than 30% of TIPS outstanding. The peak in SOMA share of TIPS market was earlier in August 2022, just after the start of quantitative tightening in June 2022. Outside of another period of quantitative easing, we expect the Fed's share of the TIPS market to continue to decline as the Fed shifts the duration of their portfolio lower by increasing their bills share.

**Figure 28: End-user demand for TIPS at auction has been increasing over the past two decades, driven by investment managers**

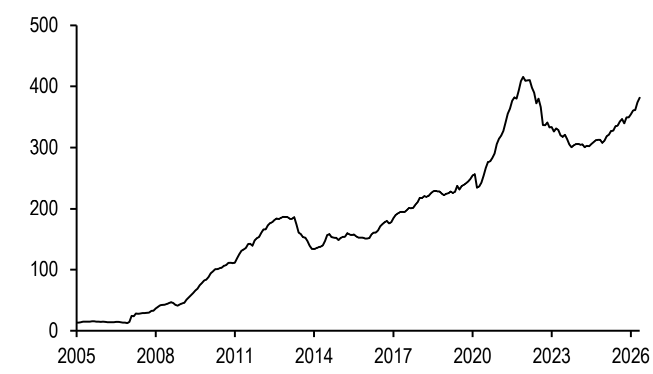
Allocation to dealers, investment funds, and foreign investors, 3-month moving average across all TIPS auctions; %




Source: US Treasury

**Figure 29: Inflation-protected mutual funds grew sharply after a series of inflation shocks**

Total net assets for inflation-protected domestic bond funds; $bn




Source: EPFR

### Our model for TIPS breakevens

**Government bond markets and nominal yields:** The dominant driver of real yields is nominal yields, with real yields tending to increase when nominal yields increase and vice-versa. Historically, the beta between TIPS yields and nominal yields has generally been below 1, meaning that TIPS yields rise less than nominal yields in a sell-off, and decline less in a rally. As a result, breakevens typically widen in a sell-off and narrow in a rally.

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**Inflation prints and expectations:** The evolution of actual headline and core CPI inflation prints generally heavily influence expectations of future inflation. In addition, factors affecting expectations of future inflation are also important drivers of TIPS and breakevens. Near-term inflation dynamics also determine the carry offered by TIPS, given their indexation mechanics. Commodity prices tend to drive headline inflation and thus heavily influence front-end TIPS. When the month-on-month change in the CPI is large, so is the carry offered by TIPS, and vice versa. It is common for investors to hedge the energy exposure on a front-end TIPS or breakeven trade by selling oil or gas futures against it. We discuss energy hedging breakeven exposure in more detail in *De-energizing inflation breakevens*, 4/17/20

**Liquidity risk (VIX Index):** Measure of implied volatility that captures changes in the demand for liquidity.Technically, TIPS have the same sovereign credit risk as nominal Treasuries, since they are both obligations of the US federal government. However, in practice, TIPS often trade like risky assets, with breakevens widening (TIPS outperforming) on “risk-on” days, and breakevens narrowing (TIPS underperforming) on “risk-off” days. One explanation for this correlation is the fact that TIPS—like several other risky asset sectors—are significantly less liquid than Treasuries, as we discussed at the start of this note. Often, flight-to-quality events are also flight-to-liquidity events, so if investor risk appetite declines—even if inflation expectations are unchanged—TIPS may underperform and breakevens may narrow.

Putting this together, we present our fair value model for inflation markets in **Figure 30**: We regress 1-month forward, seasonally-adjusted breakevens and forward zero-coupon inflation swaps on the 5 variables above. Our model implies that breakevens tend to widen as commodity prices rise but with diminishing returns, such that beyond a certain point further commodity price increases can lower breakevens. Additionally, the model implies that as volatility spikes, breakevens narrow given the lower liquidity of nominal Treasuries. Furthermore, the positive beta to the slope of future Fed pricing implies that both breakevens and the Fed react to incoming firm or soft economic data. Finally, as realized inflation rises/falls relative to economists' inflation expectations, anchoring helps to offset that impact, hence the negative beta between the variable and inflation breakevens. We use fair value models to discern richness/cheapness and help inform our outlook for breakevens.

**Figure 30: Our inflation fair value model shows breakevens are largely driven by commodity prices, Fed policy expectations, risk appetite, economist's inflation expectations**

Statistics for our breakeven and zero-coupon inflation swap fair-value model* (bp), regression over the past 7 years using daily data

| | JPMCCI | | JPMCCI^2 | | 3m3m/15m3m OIS curve; bp | | VIX Index | | Economists' inflation expectations | | Intercept (bp) | R-sq | Std error (bp) | Fair value | Residual | Current |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Beta | T-stat | Beta | T-stat | Beta | T-stat | Beta | T-stat | Beta | T-stat | | | | | | |
| 5Y | 4.01 | 60.86 | -0.0060 | -52.17 | 0.32 | 35.09 | -0.43 | -7.08 | -0.05 | -3.47 | -391 | 91% | 14.6 | 268 | -43 | 225 |
| 10Y | 2.84 | 62.02 | -0.0043 | -53.44 | 0.19 | 30.14 | -0.35 | -8.24 | -0.11 | -10.68 | -220 | 92% | 10.1 | 248 | -25 | 223 |
| 30Y | 2.06 | 49.09 | -0.0032 | -43.24 | 0.10 | 16.69 | -0.21 | -5.47 | -0.12 | -12.45 | -99 | 88% | 9.3 | 235 | -17 | 218 |
| 5y5y | 1.50 | 37.01 | -0.0024 | -33.22 | 0.08 | 13.52 | -0.28 | -7.46 | -0.23 | -24.05 | 9 | 86% | 9.0 | 254 | -16 | 238 |

*Fair value model : Breakeven or zero-coupon inflation swap (bp) regressed against JPMCCI, JPMCCI-squared, 3Mx3M/15Mx3M OIS curve (bp), VIX Index, and Economists' inflation expectations (Survey of Professional Forecasters 10Yr US CPI - (0.908 + 0.277 2Yr US CPI + 0.38810Yr US CPI), %); model uses daily data over the past seven years
Source: J.P.Morgan

## Seasonality in CPI and TIPS

Headline CPI inflation tends to be strong in the first few months of the year, moderate in the middle of the year, and soften in the last quarter of the year. This comes from various seasonal factors that drive each of the CPI components—for example, energy

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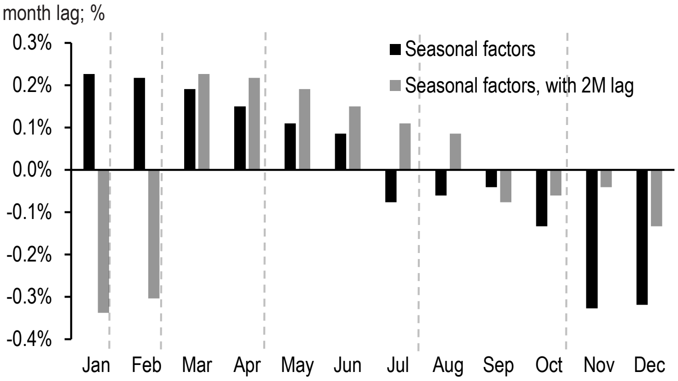
price inflation tends to firm into the peak summer travel season before moderating, while retail discounts around the holiday season in November and December tend to push apparel prices lower, etc. The BLS uses statistical methods to derive these seasonal factors, and the seasonally-adjusted CPI is often a better indicator of underlying trends in inflation. However, TIPS are indexed to the unadjusted CPI, since seasonal adjustment factors are subject to revision for up to five years after their original release. Thus, it is unsurprising that TIPS prices reflect the expected changes in CPI that normally occur at the same time each year.

Consider the relative pricing of securities maturing in different maturity months. **Figure 31** shows the additive 2025 seasonal adjustment factors from BLS by month—i.e., the difference between the m/m % changes in unadjusted and adjusted headline CPI. As mentioned, these seasonals tend to be very positive in the first few months of the year and very negative late in the year. Because of the lag in the indexation process, this means that July TIPS mature several months after positive seasonals, while January and April TIPS mature several months after negative seasonals. As a result, July TIPS trade rich relative to January and April TIPS with similar maturities. This effect is more pronounced for shorter-maturity TIPS: for example, a 0.50%-pt April-versus-July CPI differential is worth 26bp for Apr-28 and Jul-28 TIPS, but only 13bp for Apr-30 and Jul-30 TIPS (**Figure 32**).

**Figure 31: Seasonality in headline CPI...**

Difference between non-seasonally adjusted and seasonally-adjusted monthly changes in headline CPI based on 2025 BLS seasonal adjustment factors, and with 2-month lag; %




Note: Dashed lines denote when TIPS mature. These months are January, February, April, July and October. Issues maturing after positive seasonal factors, with 2M lag benefit from seasonals.
Source: Bureau of Labor Statistics, J.P.Morgan

**Figure 32: ... causes July TIPS to trade rich to April and January TIPS**

Estimated value of the seasonality difference* in return terms for TIPS of different maturity months and yield impact for TIPS impacts of different maturity years (units as indicated)

| | Jan/Apr | Jan/Jul | Jan/Oct | Apr/Jul | Apr/Oct | Jul/Oct |
|---|---|---|---|---|---|---|
| Price impact (%) | 0.12% | 0.62% | 0.64% | 0.50% | 0.52% | 0.03% |
| Yield impact, by maturity year (bp of impact) | | | | | | |
| 2028 | -7.4 | -35.2 | -34.6 | -26.4 | -26.5 | -1.2 |
| 2029 | -4.7 | -22.5 | -22.9 | -17.4 | -17.8 | -0.8 |
| 2030 | -3.4 | -16.5 | -16.8 | -13.0 | -13.4 | -0.6 |

* Based on the values in Figure 10
Source: J.P.Morgan

Similarly, seasonality matters for the pricing of a given security at different times of the year. Consider a newly issued April-maturity 5-year TIPS: principal at maturity will be scaled based on CPI changes over five full years, so seasonality has no impact. As time passes from April through January, CPI (on a lagged basis) is seasonally positive. This suggests that if you purchase the security when it has 4.5 years to maturity, for example, seasonality is expected to pressure CPI lower over the final six months before maturity, so spot prices move lower (and yields higher) to reflect this. Conversely, from January to April, CPI is seasonally negative, so spot prices rise (yields fall). Since we can quantify these effects, we often find it useful to look at seasonally-adjusted TIPS yields

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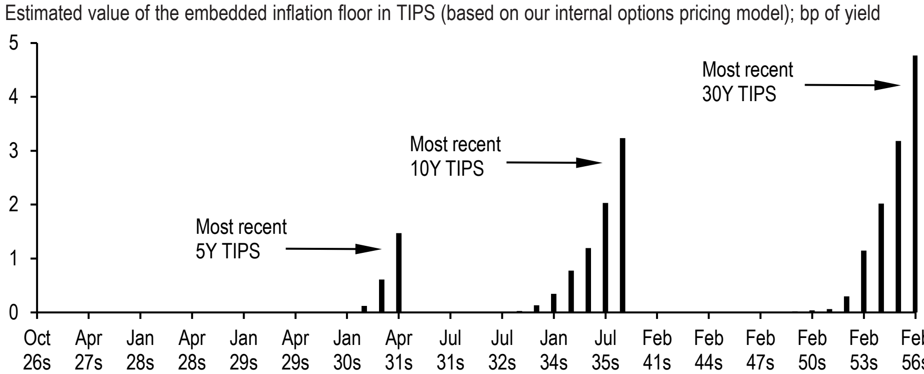
and breakevens in order to assess performance and valuations and approximate market-based measures of expectations for trend inflation (see *How do you measure, measure a year?*, 10/23/19).

**Value of the inflation floor in TIPS**

When expected future inflation is well above zero, the embedded floor on TIPS principal (floored at par at maturity) is usually worth very little, because the option is very far out-of-the-money. As **Figure 33** shows, the estimated values of the embedded floors currently range between 0-5bp of yield. These floors tend to decline in value as issues age and their index ratios rise above 1.0 (par) due to accrued inflation. The floors in recently-issued TIPS tend to be worth the most because their implied forward index ratios tend to be closest to 1.0. Additionally, TIPS with a greater time to maturity have a larger value given the time value of money for the embedded option.

Figure 33: The value of the embedded floor tends to be highest for recently-issued TIPS and increases with time to maturity

Estimated value of the embedded inflation floor in TIPS (based on our internal options pricing model); bp of yield

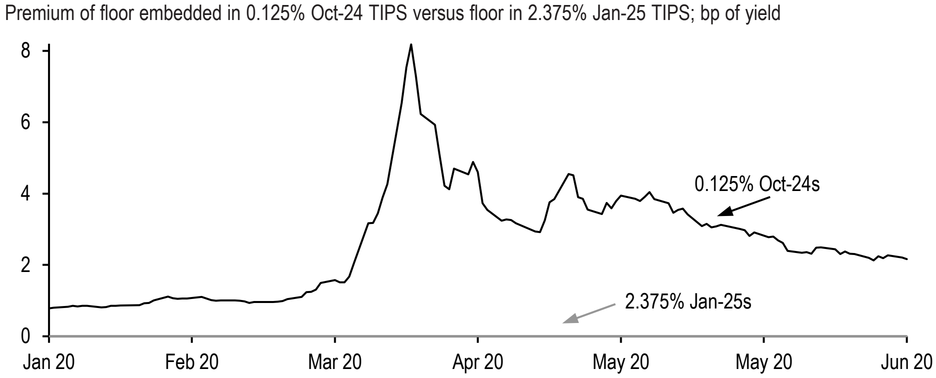



Source: J.P.Morgan

However, when expected future inflation is close to zero, the inflation floor on newly-issued TIPS can become quite valuable, as we saw during the financial crisis and COVID-19. In March 2020, inflation swap rates dropped sharply again as COVID-19 spread through the US: **Figure 34** shows that the floor value on hot-run Oct-24s TIPS rose, causing Oct-24s TIPS to outperform versus near off-the-runs with high index factors, such as the 2.375% Jan-25s, though the move was not nearly as severe as we observed in late 2008.

Figure 34: The embedded floor premiums on low-index TIPS rose in March 2020

Premium of floor embedded in 0.125% Oct-24 TIPS versus floor in 2.375% Jan-25 TIPS; bp of yield




Source: J.P.Morgan

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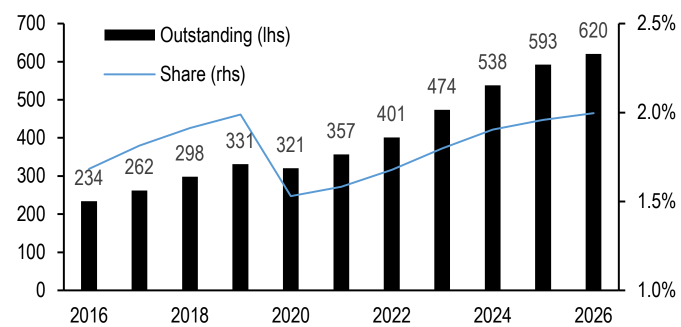
## 5. STRIPS

Unlike the previously mentioned product offerings, Treasury STRIPS, or Separate Trading of Registered Interest and Principal of Securities, are not issued or sold to investors directly. Instead, STRIPS are zero-coupon securities constructed from the separation of the coupon and principal payments of an existing coupon-bearing Treasury security. Treasury introduced the STRIPS program in 1985, though the separation of Treasury bond cash flows was popularized for its tax advantages well before the official introduction of the program, and the size of the STRIPS market now stands at $620bn, or roughly 2% of the total Treasury market (**Figure 35**). Data on the STRIPS market and stripping activity is provided on a monthly basis as part of Treasury's Monthly Statement of the Public Debt (MSPD).

When Treasury bonds are stripped of their coupon payments, they produce **Principal STRIPS** (P-STRIPS), as well as a series of **Coupon STRIPS** (Coups). The number of Coups generated depends on its final maturity: for example, a 30-year bond can be stripped into one P-STRIPS and 60 Coups, each trading with a different CUSIP. Meanwhile, the total notional amount depends on the coupon: at a 5% coupon, a stripped 30-year bond will create $100bn of P-STRIPS and $150bn of Coups (60 Coups at 2.5% semi-annual payments), but at a 3% coupon, this becomes $100bn of P-STRIPS and only $90bn of Coups (60 Coups at 1.5% semi-annual payments). Notably, Coups with the same maturity date but generated from different whole bonds are fungible (they trade with the same CUSIP), whereas same-maturity P-STRIPS are not (each has a unique CUSIP). Furthermore, STRIPS can be reconstituted into notes/bonds if a dealer or broker can obtain the P-STRIP and all corresponding Coups.

**Figure 35: The STRIPS market stands at $620bn and is a 2% share of the total Treasury debt outstanding**

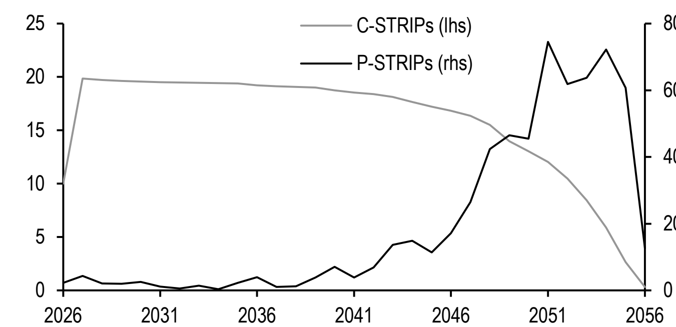
P-STRIPS outstanding (lhs;$bn), and as share of total marketable debt outstanding (rhs; %)




Source: US Treasury

**Figure 36: P-STRIPS and Coup supply tends to be concentrated in the long end and short end, respectively**

Coups and P-STRIPS outstanding by year of maturity, as of 5/30/2026; $bn on both axes




Source: US Treasury

### Sources of demand

Stripping bonds results in P-STRIPS that have a longer duration than the whole bonds themselves, due to their zero coupon structure. Moreover, because STRIPS trade at a discount to par, owning the same dollar amount of STRIPS instead of their whole bonds effectively allows for a levered duration position. Thus, these instruments are ideal for liability-driven investors (LDIs) like pension funds and insurance companies. For pension funds, STRIPS are a fitting instrument for predictably matching cash flows to future liability payments. Because durations are relatively longer than their unstripped counterparts, STRIPS allow for stronger hedging against liabilities facing interest rate risk. Moreover, because they typically trade at a deep discount to par, STRIPS allow investors to extend duration with comparatively low levels of capital, making them highly capital efficient investments for interest rate exposure.

Due in part to their popularity amongst the LDI community, stripping activity is concentrated at the long-end of the curve. To a lesser extent, stripping activity also occurs at the front end of the Treasury curve, as short-dated STRIPS are often used as a cash management tool (given their singular cash flow payout), particularly for defeasance purposes. For instance, a bond issuer may use some of the issuance proceeds to purchase STRIPS in order to match the size and timing of the bond's cash flow payments. At the same time, the process of stripping a long-duration bond creates an annuity of Coups, resulting in greater Coup supply in the belly of the curve. Elevated stripping activity and high coupons have created a surfeit of C-STRIPS, which has placed upward pressure on Coup/P spreads in recent years (see "STRIPS Relative Value" below) (**Figure 36**).

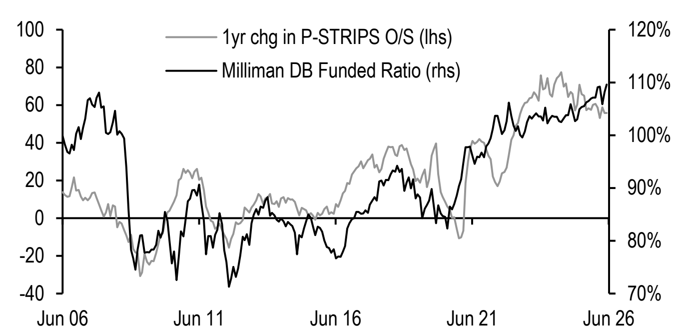
### Drivers of demand

There are a variety of drivers of LDI demand for STRIPS that have helped explain fluctuations in stripping activity. However, there is one driver in particular that is preeminent: pension funded ratios. The funded ratio of a pension plan is equal to the value of its assets divided by the present value of its future liabilities. Accordingly, the funded ratio can broadly be looked at as a metric of a pension fund's ability to cover its financial commitments.

A defined benefit (DB) pension plan requires employers to make contributions to a pooled fund dedicated to paying out future retirement benefits. The pension fund is then responsible for deciding how to allocate that net contribution into various securities/ asset classes to ensure it maintains the capital necessary to fulfill its current and future obligations. Prior to the GFC, asset allocation leaned heavily towards equities, but the decline in yields and equities exposed this asset/liability mismatch and drove pension funded ratios well below 100%. In the decade following the GFC, as risk assets slowly recovered and yields climbed from their post-crisis lows, the funded gap closed and pension funds reallocated into fixed income in order to immunize funding risks. Demand for STRIPS rose as pension funds added long-duration assets in an effort to better match the duration of their long-dated liabilities, and hence minimize the impact of interest rates on their portfolios over time (**Figure 37**).

**Figure 37: DB pension funded ratios have been a historical driver of stripping activity, but their importance has declined in recent years**

Milliman PFI funded ratio (%, lhs) versus 1-year change in P-STRIPS outstanding ($bn, rhs)




Source: J.P. Morgan, Milliman PFI

**Figure 38: ...as pension funds have increasingly allocated into fixed income and the volatility of their surpluses has fallen**

NISA Pension Risk Surplus Index*; %


*See NISA Pension Surplus Risk Index
Source: NISA

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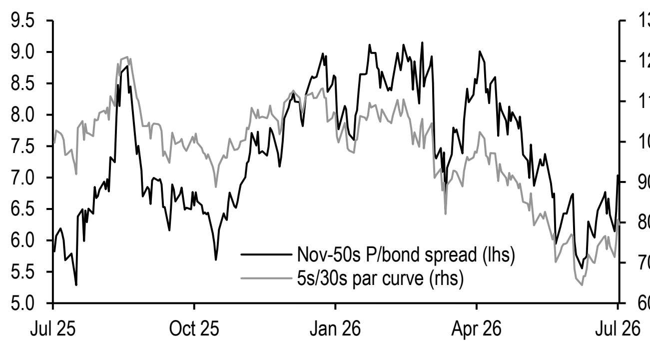
However, this relationship has weakened in the post-pandemic environment. First, the largest DB pension funds have been in surplus for the past four years, and their fixed income allocation has risen above 50%, warranting a slower pace of demand for risk-free longer-duration assets compared with the torrid pace observed in recent years. As a result, the expected volatility of pension funded ratios has fallen significantly (**Figure 38**). Second, given that most corporate DB plans closed to new entrants over a decade ago, the duration of this liability should shorten over time, meaning that what demand persists should also roll down the curve to the intermediate sector in the coming years.

### STRIPS relative value

P-STRIPS can at times offer relative value opportunities versus whole bonds. Given that STRIPS have a longer duration than their similar-maturity whole bond counterparts, trading P-STRIPS against the whole bond can be thought of as an alternative way to position for changes in curve shape. For instance, the spread between the Nov-2050s P-STRIPS and the 1.625% Nov-2050s is highly correlated to moves in the 5s/30s Treasury curve (**Figure 39**). This relationship can also be used to identify sources of relative value across P/bond spreads. **Figure 40** shows outright P/bond spreads and residuals after regressing against the 5s/30s Treasury par curve for a handful of STRIPS at the long end: a positive residual suggests that the P-STRIPS is trading cheap to the whole bond, after adjusting for the broader Treasury curve, and vice versa.

**Figure 39: The P/whole bond spread is driven by the shape of the long end of the curve...**

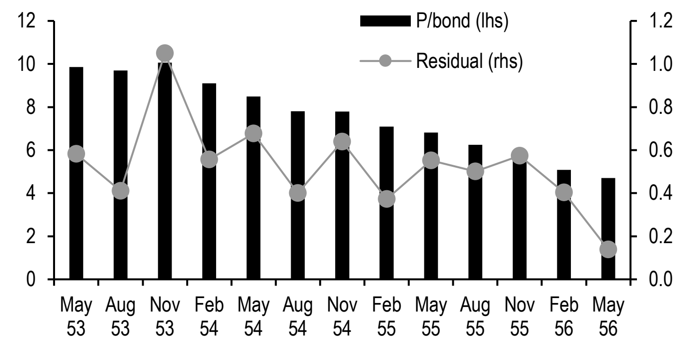
Nov-50s P-STRIPS/1.625% Nov-50s spread (lhs) versus 5s/30s Treasury benchmark curve (rhs); bp




Source: J.P. Morgan

**Figure 40: ...and the residual from regressing P/bond spreads on the slope of the curve is a useful tool for identifying relative value**

Outright P/bond spreads and residual* after regressing against 5s/30s Treasury par curve for select STRIPS at the long end; regression over a 6-month horizon; bp on both axes




*As of 7/16/26
Source: J.P. Morgan

Additionally, there are differences in valuations between similar-maturity Coups and P-STRIPS, known as the Coup/P spread (**Figure 41**). Because most stripping activity tends to occur at the long end (as discussed earlier), the lion's share of P-STRIPS outstanding is generally concentrated at the long end of the curve, whereas the stripping process creates a surplus of Coups in the belly of the curve, where there are fewer natural buyers. Given the differences in stripping supply and demand in various sectors of the curve, greater net stripping activity tends to widen Coup/P spreads and vice versa, particularly in the intermediate sector. As a less liquid, balance sheet intensive sector of the Treasury market, Coup valuations are also impacted by broader risk appetite and liquidity conditions and we have found that Coup/P spreads tend to widen when implied volatility rises. Together, these factors explain roughly 80% of the variation in Coup/P spreads over the past five years (**Figure 42**).

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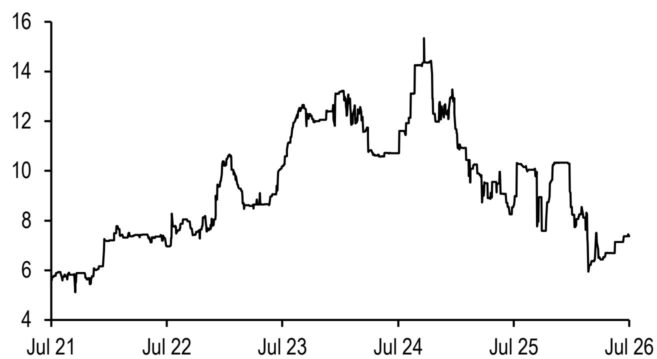
**Figure 41: Coup/P spreads have tightened to near the tight end of their post-COVID ranges...**

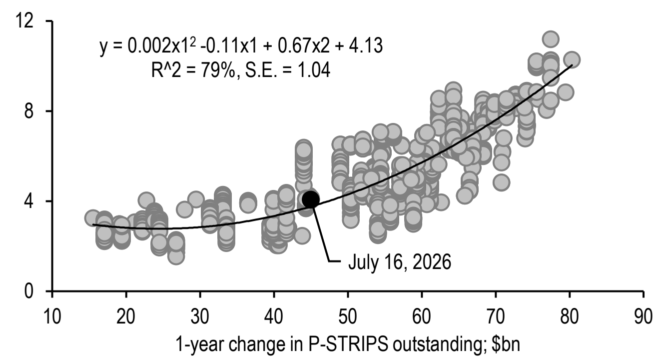
Average Coup/P spread for all STRIPS in 2042-2052 sector; bp




Source: J.P. Morgan, U.S. Treasury

**Figure 42: ...and the narrowing has been well explained by the pace of stripping activity and the level of implied volatility**

Average Coup/P spreads for the 2042-2052 sector regressed on 12-month change in P-STRIPS outstanding ($bn) and 1yx10y swaption implied volatility (bp/day), regression over the last 5 years; bp



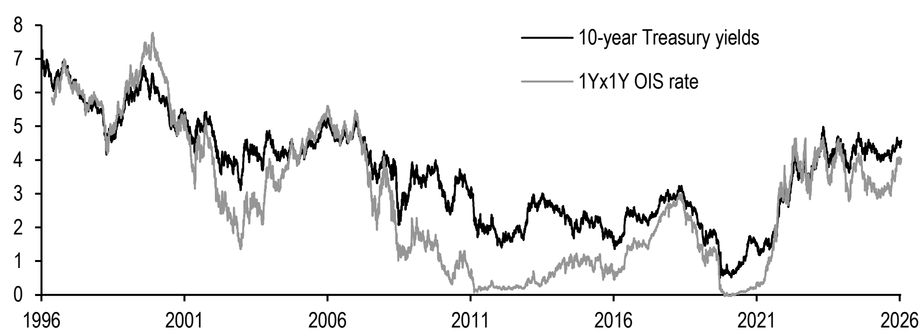

Source: J.P. Morgan, U.S. Treasury

# Key drivers of yields

Over the long term, the level of Treasury yields is most influenced by the expected stance of monetary policy: **Figure 43** shows that 10-year Treasury yields have tracked the 1Yx1Y OIS rate quite closely over the past thirty years, and this factor explains roughly 87% of 10-year yield variation over this period. Along the curve, Fed policy expectations are clearly more influential on shorter-maturity Treasuries: by comparison, the 1Yx1Y OIS rate explains 95% and 71% of the variation in 2-year and 30-year yields, respectively, over the same period. Since future adjustments to the Fed funds rate are dictated by developments in inflation and unemployment, long-end yields incorporate a more medium-term outlook on macroeconomic variables.

**Figure 43: Policy rates have been the primary driver of Treasury yields over the longer term**

10-year Treasury yields and 1Yx1Y OIS rates; %




Source: J.P. Morgan

Beyond their expectations for Fed policy, investors require extra compensation for the interest rate risk and funding lockup inherent in holding long-duration Treasury bonds. This extra yield compensation is referred to as term premium, which increases in maturity and is closely related to the slope of the curve. We have found term premium is influenced by the balance between supply and demand in the Treasury market as well as the volatility of interest rate and inflation expectations (see *In the eye of the beholder*,

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9/7/23). Indeed, various measures of term premium have crept higher in recent years, normalizing back to pre-GFC levels, likely buoyed by the aforementioned demand shift and increased inflation and policy uncertainty (**Figure 44**).

We develop valuation frameworks for various maturity points along the Treasury curve, as well as the shape of the yield curve, using regression analysis based on empirical fundamental and technical factors. These models help drive our expectations for the future path of yields, based on the outlook for the underlying factors, and inform our views around whether Treasuries appear rich or cheap relative to their drivers. We monitor how well the factors explain yield changes, and periodically modify the frameworks. We review our current fair value models over the following sections.

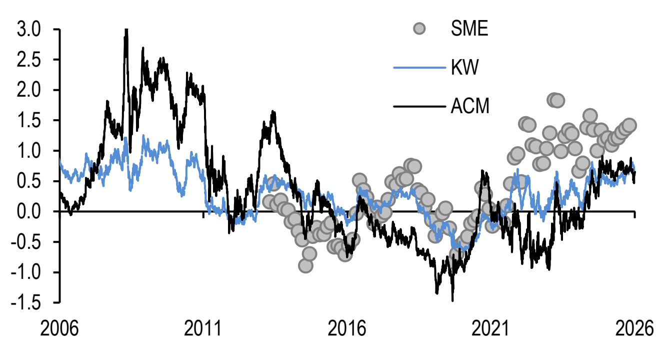
## 10-year fair-value model

As highlighted previously, the core of our 10-year fair value model rests on medium-term Fed policy expectations, as proxied by the 1Yx1Y OIS rate, as well as variables that reflect medium-term expectations for macroeconomic outlook. We augment Fed policy expectations with market-based medium-term inflation expectations derived from 5Yx5Y seasonally-adjusted TIPS breakevens (see *Dropping anchor: Why five year forward five-year inflation has stayed rangebound*, Harry Downie, 6/9/26) as well as the J.P. Morgan US growth Forecast Revision index (FRI) to reflect shifts in growth expectations (see *Know thyself: Evaluating and using J.P. Morgan economic forecasts*, Lupton, Kasman, and Loeys, 9/23/14).

While market-based expectations of medium-term Fed policy, inflation, and growth are a good starting point for our models, we find that other policy levers have also been significant drivers over time. The large-scale purchases of Treasuries and MBS that have led to the meteoric growth in the Fed's balance sheet over the past two decades have had a significant and durable impact on Treasury yields. Indeed, we have found that 10-year Treasury yields tend to decline by 10bp for every 1% increase in the size of the Fed's balance sheet as a share of US GDP, holding other factors constant. Furthermore, the addition of this factor in our model incorporates the gradual demand shift away from price-insensitive investors underway in Treasury markets.

**Figure 44: Various measures of term premium have been increasing in recent years, normalizing to their pre-GFC levels**

ACM, KW, and Survey of Market Expectations* measures of 10-year Treasury term premium*; %




*For SME we average 10-year zero coupon yield from Treasury par curve for the two weeks ahead of SME submission dates and subtract 10-year average expectation of the Federal funds rate
Source: Federal Reserve Board, Federal Reserve Bank of New York, J.P. Morgan

**Figure 45: We find that medium-term Fed, inflation, and growth expectations, as well as Fed balance sheet size and policy uncertainty explain much of the variability in intermediate yields**

J.P. Morgan 10-year Treasury fair-value model*; units as indicated

| Variable | Coefficient | T-stat | Current |
|---|---|---|---|
| Intercept | 7.3 | 11.6 | |
| 1y1y OIS rate; % | 0.66 | 87.4 | 4.0 |
| JPM US FRI; % pts | -0.04 | -5.8 | 92.6 |
| 5Yx5Y TIPS breakevens; % | 0.12 | 2.5 | 2.0 |
| Fed B/S share of GDP; % | -0.10 | -33.8 | 21.0 |
| Tariff dummy variable | 0.10 | 6.2 | 0.0 |
| R-squared; % | | | 97.3 |
| Standard Error; bp | | | 14 |
| Residual; bp | | | -15.6 |

*Regression over the past five years. Tariff dummy variable set to 1 between 4/2/25 and 2/26/26
Source: J.P. Morgan, Federal Reserve Bank of New York, US Treasury

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Finally, we include a dummy variable in our framework to capture trade policy uncertainty, which increased the fair value of intermediate yields in the wake of President Trump's "Liberation Day" announcements in April 2025, but which has been somewhat attenuated since the Supreme Court struck down the IEEPA tariffs in February 2026. Overall, when we add these factors, our model explains 97% of the variation in 10-year yields over the past 5 years (**Figure 45**).

## 5s/30s curve fair-value model

Our framework for valuing the curve utilizes the same factors as detailed above, except for the JPM growth FRI. **Figure 46** illustrates that medium-term Fed policy expectations remain the most important driver of curve shape, which is theoretically providing the greatest influence over the short-end of the curve. Meanwhile, Fed balance sheet size is far more impactful on curve slope than outright yield levels, which is likely providing the greatest influence over the long end. Indeed, we find that the 5s/30s curve flattens by roughly 2bp for every 1% increase in the size of the Fed's balance sheet as a share of US GDP, holding other factors constant. On balance we find these factors explain more than 92% of variation in the slope of the 5s/30s curve.

**Figure 46: Fed policy expectations and Fed balance sheet size are the two most significant factors in determining the slope of the Treasury curve**

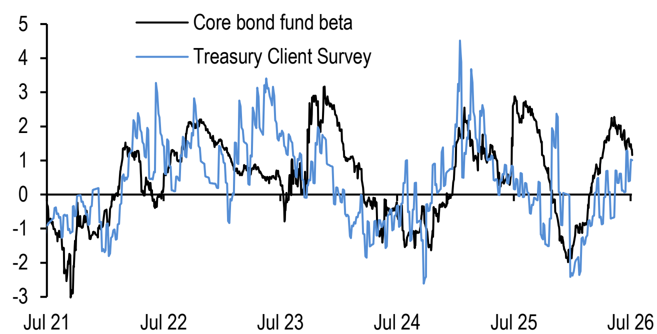
J.P. Morgan 5s/30s Treasury fair-value model*; units as indicated

| Variable | Coefficient | T-stat | Current |
|---|---|---|---|
| Intercept | 291.7 | 37.8 | |
| 1y1y OIS rate; % | -32.5 | -72.0 | 4.0 |
| 5Yx5Y TIPS breakevens; % | 8.7 | 2.5 | 2.0 |
| Fed B/S share of GDP; % | -5.8 | -67.6 | 21.0 |
| Tariff dummy variable | 19.8 | 17.7 | 0.0 |
| R-squared; % | | | 91.9 |
| Standard Error; bp | | | 11.8 |
| Residual; bp | | | 23.1 |

*Regression over the past five years. Tariff dummy variable set to 1 between 4/2/25 and 2/26/26
Source: J.P. Morgan, Federal Reserve Bank of New York, US Treasury

**Figure 47: Swings in Treasury demand can provide tactical trading strategies when positioning becomes stretched**

Rolling 1-year z-score of J.P. Morgan Treasury Client Survey Index* and corre bond fund beta to 10-year Treasury yields**




*(Longs + Neutrals)/(Shorts + Neutrals)
**The core bond fund index is comprised of the 25 largest actively-managed US core bond funds by AUM. Model is a 3-month regression of daily excess returns on the bond fund index versus daily changes in 1) 10-year US Treasury yields, 2) 5s/30s Treasury curve, 3) JULI spread to Treasury, 4) MBS current coupon Treasury OAS, and 5) 3Mx10Y swaption volatility
Source: J.P. Morgan, Bloomberg Finance L.P.

We frequently refer to these fair-value models in our research, and we publish reports on these models on our daily *US Cash Interest Rate Products Analytics Package*, available on the US Rates Strategy page on J.P. Morgan Markets.

Away from signals from our fair value models over long horizons, swings in demand for Treasuries can have a significant impact on yield levels over the short term —for example, pushing yields lower if investors in aggregate reach for securities as a source of carry or sending yields higher if speculative investors build substantial short positions. When investor positions deviate sharply from average levels, they portend a reversal in yields, and we have found that the Treasury Client Survey Index can be used to trade duration in a contrarian fashion (see *Survey says: using the Treasury Client Survey to predict rates moves*, 7/21/23). As such, we monitor various positioning metrics to inform tactical trading strategies (**Figure 47**, see below).

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## Estimating investor positions in Treasuries

We track a number of positioning metrics, which are updated regularly in the Appendix of our *US Treasury Market Daily* publication and include the following:

**Treasury Client Survey**: This survey is conducted on the first business day of the week. Clients are asked if they are currently short, neutral, or long duration relative to their benchmark. If the client does not have a benchmark, the outright duration position is indicated. For the overall survey, 40-60 clients are surveyed, with an approximate breakdown of 60% real money, 25% spec accounts, and 15% central bank/sovereign wealth funds. Our Treasury Client Survey Index attempts to quantify how pervasive positioning is, by adjusting each longs and shorts by the share of neutrals in the overall survey (U.S. Treasury Client Survey).

**CFTC non-commercial positions**: Released on the Friday of each week with a breakdown of Treasury and Eurodollar futures positions as of the previous Tuesday, CFTC's Commitments of Traders report divides reportable traders into "commercial" and "noncommercial" categories. The latter category generally represents positions of speculative traders (available here).

**CTAs**: This estimate is the partial beta of CTA excess returns over cash (based on the NEIXCTA Index from Bloomberg) with respect to the JPM US GBI 7-10Y bond index. The model estimates CTA exposures by regressing CTA excess returns on excess returns of a number of variables over a 1-month horizon: JPM 7-10Y US GBI, JPM Global Bond Index ex-US, S&P 500 index, JPM cash index, Goldman Sachs Commodities Index (see *US Treasury Market Daily*, 9/28/17).

**Active Core Bond Fund Managers**: This is the partial beta of US active core bond fund excess returns with respect to 10-year US Treasury yields. Core bond funds typically provide investors with a broad exposure to the US IG bond market. We track daily excess returns of the 25 largest (by AUM) core bond funds relative to a benchmark and compute the asset value-weighted average excess return with respect to our GABI. This is then regressed on a number of variables over a 3-month horizon: 10-year Treasury yields, 5s/30s Treasury curve, JULI spread to Treasuries, MBS current coupon Treasury OAS, and 3Mx10Y swaption volatility (see *US Treasury Market Daily*, 5/14/26).

**Macro Hedge Fund**: This is a partial beta of macro HF excess returns (based on the HFRXM Index from Bloomberg) with respect to the JPM US GBI 7-10Y bond index. We estimate macro hedge fund exposures by regressing excess returns over cash on the following factors over a 6-week horizon: JPM 7-10Y US GBI, JPM Global Bond Index ex-US, S&P 500 index, JPM cash index, Goldman Sachs Commodities Index, MSCI World ex-US (see *US Treasury Market Daily*, 7/14/15).

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# Relative value

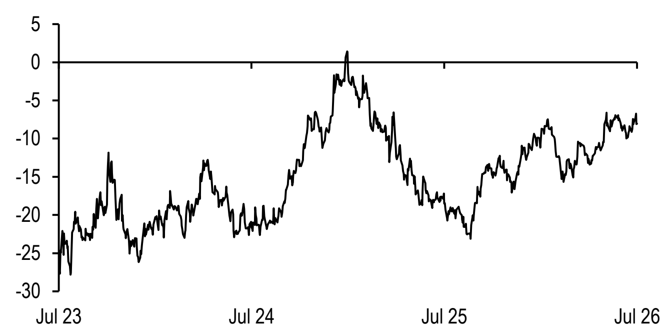
Much of our discussion thus far has been focused on valuation frameworks for Treasury yields and the shape of the yield curve. To take it one step further, beyond outright value, we consider relative value, or the richness or cheapness of assets after adjusting for their underlying drivers. Macro relative value involves broad curve relative value, or discerning the relative richness or cheapness of a sector along the Treasury curve, while micro relative value applies to security selection in Treasury securities.

## Macro relative value

As the name implies, macro relative value is the practice of discerning value for a given sector of the Treasury curve, after adjusting for its fundamental drivers. In the simplest sense, we like using Treasury butterflies to isolate the value in specific sectors along the curve. **Figure 48** displays the 2s/5s/10s Treasury butterfly, and shows 5-year Treasuries have underperformed the 50:50 weighted combination of 2- and 10-year Treasuries by roughly 10bp over the past year, and are now trading near their cheapest levels since April 2025. On the surface this indicates the 5-year sector is cheap along the curve but ignores the drivers of this spread. **Figure 49** shows this same butterfly regressed on the level of rates and the shape of the curve. In this case we regress the butterfly on the body of the butterfly (5-year yields) and the slope of the wings (2s/10s curve). The regression shows these factors explain more than 60% of the variation in this butterfly over the past 3 years, with the butterfly cheapening by roughly 1bp for every 10bp increase in 5-year yields as well as 10bp steepening of the 2s/10s curve. Finally, adjusting for these drivers, this shows the 5-year sector appears fairly valued relative to the wings.

**Figure 48: Treasury butterflies can be useful in discerning value in specific sectors along the curve...**

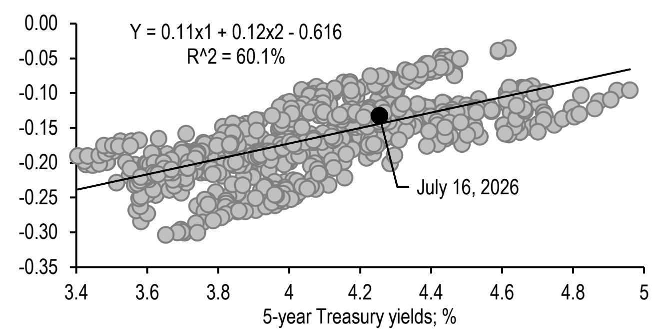
2s/5s/10s Treasury butterfly; bp




Source: J.P. Morgan

**Figure 49: ...particularly when regressed on the level of rates and the shape of the curve**

2s/5s/10s Treasury butterfly regressed on 5-year Treasury yields (%) and 2s/10s Treasury curve (%); regression over the last three years; %




Source: J.P. Morgan

More generally, we can apply this level- and curve-neutral analysis across the curve: **Figure 50** displays regression statistics for a number of Treasury benchmark butterflies. The table shows that the level of rates and slope of the curve explain much of the variation in these butterflies, and the explanatory value is relatively higher for wide butterflies than more narrow spreads. Therefore, we think this is a robust measure for evaluating Treasury value. We offer a "US Treasury Butterfly Report" as part of our daily *US Cash Interest Rate Product Analytics Package*, available on the US Rates Strategy page on J.P. Morgan Markets.

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**Figure 50: The level of rates and the slope of the curve are the primary two drivers of most wide Treasury butterflies**

Regression statistics for various Treasury butterflies (%) regressed on the level of rates of the body of the butterfly (%) and the slope of the curve of the wings of the butterfly (%), regression over the last 6 months; units as indicated

| | | | Level (bp) | | Curve (bp) | | |
|---|---|---|---|---|---|---|---|
| Butterfly | Current | R^2 (%) | Beta | T-stat | Beta | T-stat | Resid (bp) |
| 2s/3s/5s | -1.6 | 88% | 6.59 | 24.3 | 0.0 | 0.0 | 0.2 |
| 2s/5s/10s | -8.1 | 65% | 13.39 | 14.3 | 0.1 | 8.2 | 0.3 |
| 2s/5s/20s | -34.0 | 62% | 14.15 | 7.7 | 0.0 | -0.5 | 0.0 |
| 2s/5s/30s | -33.8 | 81% | 19.64 | 11.2 | 0.0 | -0.7 | -0.7 |
| 3s/5s/10s | -10.0 | 74% | 6.01 | 8.0 | 0.0 | -1.6 | 0.1 |
| 3s/5s/30s | -35.7 | 92% | 8.00 | 6.3 | -0.2 | -10.8 | -0.6 |
| 3s/7s/10s | 2.7 | 91% | 6.53 | 12.8 | 0.3 | 29.4 | -0.4 |
| 3s/7s/30s | -23.0 | 60% | 11.08 | 8.7 | 0.0 | 2.0 | -1.5 |
| 5s/10s/30s | -12.2 | 18% | 2.81 | 3.5 | 0.0 | 5.1 | -1.2 |
| 10s/20s/30s | 26.2 | 50% | 9.66 | 8.6 | 0.3 | 11.1 | -1.0 |

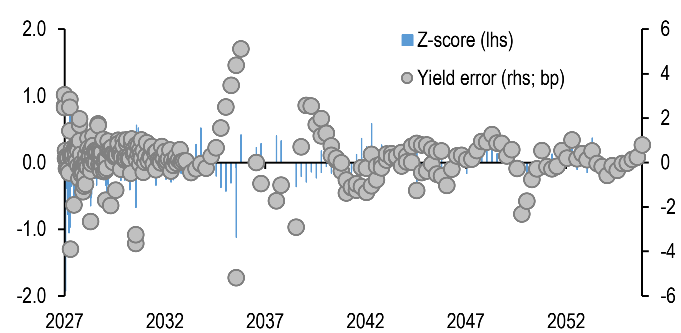
Source: J.P. Morgan

## Micro relative value

While macro relative value is focused on a sector-by-sector analysis of the curve, micro relative value is centered on estimating the value of specific Treasury securities. A yield curve is needed to effectively estimate the richness or cheapness of these securities, and we use our par-fitted Treasury curve as the basis for evaluation (see *The (par) curves they are a-changin': Making enhancements to our Treasury & TIPS par curves*, 7/23/24). The most basic output is the yield error—the difference between observed yield on a particular Treasury security and the yield derived from our par fitted curve for the same security. A positive yield error implies the bond appears cheap relative to the curve, while a negative yield error implies the bond is rich (**Figure 51**).

**Figure 51: Yield errors to our par curve can be a useful micro relative value measure**

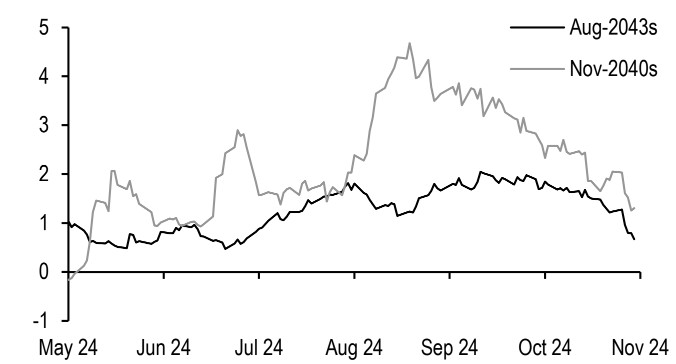
Yield errors relative to the J.P. Morgan par curve (rhs; bp) versus 3-month Z-score of yield errors (lhs);




Source: J.P. Morgan

**Figure 52: The Nov-40s cheapened relative to the par curve as volatility increased in the summer 2024, relative to the more liquid Aug-2043s**

Yield error for the 3.625% Aug-2043s and the 1.375% Nov-2040s; bp




Source: J.P. Morgan

**Figure 52** shows the yield errors for the 3.625% Aug-2043s and the 1.375% Nov-2040s, both deep off-the-runs of similar durations around 13.5 years with a similar 30-40% share held by the Fed's SOMA portfolio. Despite these similarities, these two off-the-runs traded very differently during the August 2024 bout of volatility associated with a surprisingly weak labor report and carry trade unwinds. The Aug-43s richened to the par

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curve in the immediate wake of the report, and remained within a stable range thereafter, whereas the Nov-40s underperformed sharply, cheapening 4bp relative to the par curve. Notably, the Aug-2043s were original issue 30-year bonds of $42bn size, whereas the Nov-2040s were original issue 20-year bonds of $86bn size. Indeed, the Nov-2040s were part of the initial bumpy re-introduction of the 20-year sector in 2020, wherein Treasury rolled out much larger sizes than primary dealers' recommendations when funding needs grew during the onset of the pandemic. Thus, it is not surprising that given its large size and difficult subscription, liquidity in these OI 20-year bonds has been historically poor when compared to neighboring tenors.

While yield errors are clearly useful for finding dislocations on the Treasury curve, we prefer using our rich/cheap measure, which is the spot yield error for a given bond less the three-month moving average of the yield error—effectively adjusting for any idiosyncratic richness or cheapness of the security. Our "Treasury Carry, Roll, and Relative Value Report," which shows yield errors and rich/cheap measures for all outstanding coupon Treasuries, can also be found in the daily US Cash Interest Rate Product Analytics Package, available on the US Rates Strategy page on J.P. Morgan Markets.

## Market structure and liquidity

The cash Treasury market operates in an over-the-counter, decentralized format. The market can be divided into three major subcomponents: the interdealer broker (IDB), dealer-to-dealer (DTD), and dealer-to-client (DTC) segments. In the IDB market, certain brokers and principal trading firms intermediate between dealers, with most trades occurring on electronic platforms in on-the-run securities, and some off-the-run activity on manual-assisted platforms. In the DTD market, which is the smallest of the three segments, dealers trade directly with each other. Finally, in the DTC market, dealers facilitate the demands of institutional end-users, and trading frequently takes place in off-the-run securities over electronic request-for-quote, direct streaming, and voice/message platforms.

As is evident, primary dealers play a centralized role in Treasury market functioning. As we discussed earlier, the role of the primary dealer community has changed over the past two decades. Consolidation before and during the financial crisis reduced the number of primary dealers from 27 in 2001 to just 17 in 2008, but the number of dealers has steadily increased following the GFC and now stands at 26 (**Figure 53**). While the number of dealers has recovered, their holdings have not; dealer positions in Treasuries have grown on an outright basis but have not been able to keep pace with the rapid growth of the Treasury market, and only represent 1.5% of total outstanding Treasuries, compared to 4% on the eve of the GFC (**Figure 54**).

Certainly, regulatory developments, including the US Supplementary Leverage Ratio (SLR), first proposed in 2013, and the US G-SIB surcharge, first proposed in the fall of 2014, have limited the accumulation of inventory, and leaned heavily on the principal trading model. The consistent theme running through these regulatory developments is the high cost of taking and holding positions (assets) and financing these positions (liabilities) on bank balance sheets. Accordingly, banks became constrained not just by the composition of their balance sheets (i.e. risk-weighted assets), but also by the sheer size. In addition to regulation, at least a portion of this reduced risk tolerance is self-inflicted in nature: the experience of the financial crisis has led dealers to pare back on

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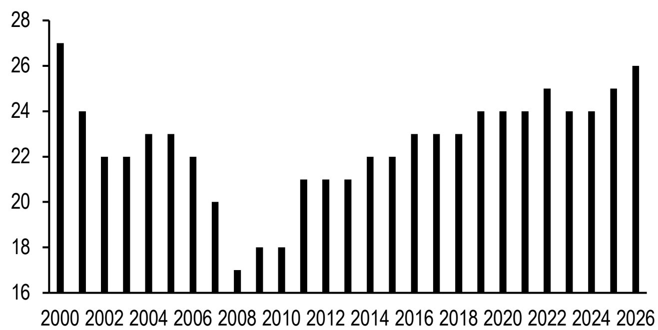
the risk profile of their trading books, and risk management and control frameworks at banks have become more sophisticated in the post-crisis era, resulting in lower risk appetite.

**Figure 53: The group of primary dealers has risen to their highest number since 2000...**

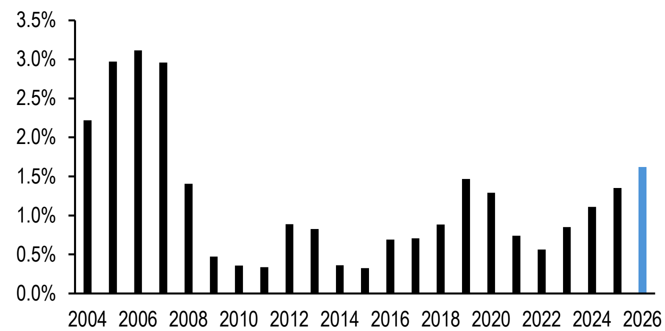
Number of Treasury primary dealers at year-end*




*2026 through July 2026
Source: Federal Reserve Bank of New York

**Figure 54: ...though dealer positions remain lower on a relative basis than prior to the GFC, even if they have recently been increasing**

Aggregate magnitude of primary dealer positions in Treasuries as a share of marketable Treasury debt outstanding*;




*2026 through July 2026
Source: Federal Reserve Bank of New York, US Treasury

However, dealer inventory has been steadily climbing as a share of the Treasury market in recent years with limited evidence of market stress, in what we believe reflects increased dealer capacity to warehouse risk. This in part is likely tied to the deregulatory push from the current administration. Last year US banking regulators reduced the enhanced SLR standard for G-SIBs and their depository institution subsidiaries to 50% of their method 1 surcharge (see *SLR reform update*, Ipek Ozil, 11/12/25). More recently, the long-awaited Basel III endgame and G-SIB surcharge reform proposals were released, which would likely reduce aggregate CET1 capital requirements across the banking system. The easing of leverage requirements and release of capital could facilitate market-making activities and improve liquidity (see *Jokes Aside: GSIB and Basel III Endgame Update*, Teresa Ho and Ipek Ozil, 4/1/26).

Other ancillary factors have also supported increased dealer participation over the past year. The lift on Wells Fargo's asset cap in the summer of 2025 has allowed the bank to expand its balance sheet for the first time since 2018, likely augmenting primary dealer activity. The Fed's decision to conduct Reserve Management Purchases (RMPs) of T-bills in an effort to maintain an ample level of aggregate banking reserves has provided support to funding markets, reducing costs associated with holding Treasury positions. Furthermore, Treasury's liquidity support buybacks have likely eased dealer warehousing costs by allowing them to offload expensive and illiquid deep off-the-run securities.

While all of these factors have aided in the resumption of dealer participation, their footprint is still diminished relative to prior to the GFC, allowing new entrants into the Treasury market. In the wake of the flash crash of October 2014, the Inter-Agency Working Group (IAWG) for Treasury Market Surveillance (which comprises of the SEC, Fed, Treasury, New York Fed, and CFTC) released a Joint Staff Report which highlighted that principal trading firms (PTFs) accounted for more than half of the trading volumes on the major IDB electronic trading platforms. Many of these PTFs engage in algorithmic and high-frequency trading strategies (HFT), through which

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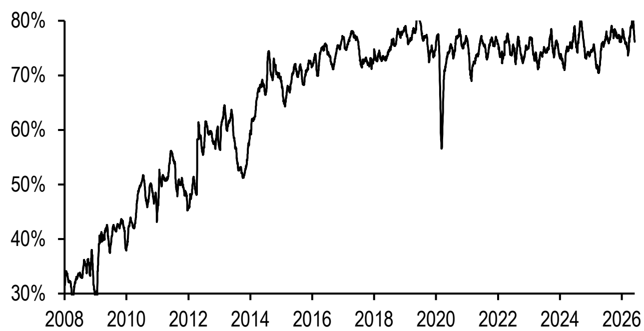
algorithmic activity reacting on a timescale of micro-seconds has come to dominate liquidity provision in the interdealer market for Treasuries (**Figure 55**, see *Far from the shallow now? Liquidity provision by high frequency participants in US rates*, 4/12/19). However, as the exhibit shows, this share can be variable and is very sensitive to volatility: the HFT share tends to recede rapidly in periods of elevated volatility, most dramatically in March 2020. On margin, their dominant presence in liquidity provision can be procyclical in nature, contributing to improved liquidity in times of low volatility, but exacerbating deterioration in liquidity in times of heightened volatility, like what was witness in the spring of 2020.

**Figure 55: High-frequency algorithmic activity dominates liquidity provisioning in the interdealer market**

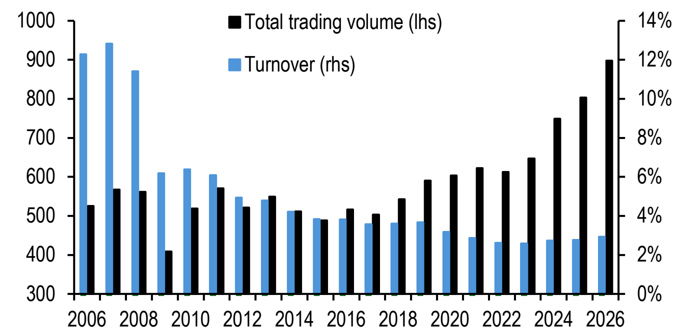
The fraction of 10-year Treasury market depth provided by high frequency activity*; 1-month moving average




*Defined as passive limit orders that are reacting within 0.1 second to another event in the CLOB
Source: J.P. Morgan, BrokerTec

**Figure 56: Trading volumes have increased 60% since prior to the pandemic and turnover has stabilized at historically low levels**

Average daily Treasury trading volume (lhs; $bn) versus daily Treasury market turnover (rhs; %)




Source: Federal Reserve Bank of New York, US Treasury

## Treasury market liquidity

In addition to their strong credit, one of the central and most attractive features of Treasury securities is their robust liquidity, which in part is tied to the ease in which investors can transact and convert their holdings into cash. This reflects the fact that the Treasury market is the largest and deepest fixed income market in the world; accordingly, it is the risk-free benchmark for pricing other financial assets. Yet the growth in the size of the market over the past two decades has outstripped participation as measured through daily trading volumes, and consequentially turnover has declined to a fraction of its pre-GFC levels. That said, turnover has stabilized at low levels amid an uptick in trading volumes since the pandemic (**Figure 56**). This uptick has likely been the result of the shifting demand backdrop from more price insensitive and buy-and-hold end-users towards more price sensitive and thus more active investor base.

Treasury trading volumes are concentrated in on-the-run coupon securities, and of those the more liquid tenors of the 2-, 5-, 10-, and 30-year maturities. As such, when assessing overall market liquidity, we prefer inspecting duration-weighted market depth across these tenors during the New York morning trading session, which is the most active period of the day. IDBs operate central limit order books (CLOBs) to collect and consolidate trading interest from primary dealers and PTFs, recording how much notional participants were willing to transact at the best (bid/ask) price, and at each tick tier behind that. **Figure 57** shows the aggregate sum of the three best bids and offers by tick queue position across these tenors, averaged during this time slot. This depth dealers are comfortable showing on screens has fluctuated over the years, falling precipitously during the onset of the COVID-19 pandemic, but has steadily increased in

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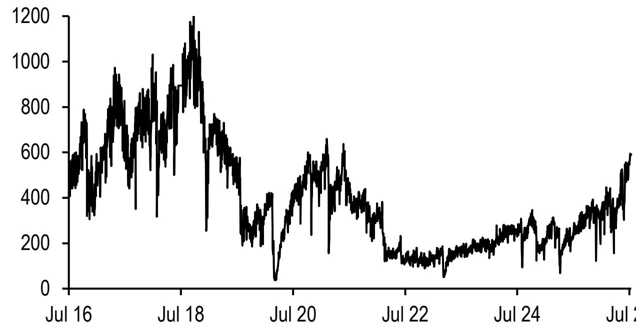
recent years since their nadir during the 2023 banking tumult. This improvement has been supported by declining volatility, which peaked during the 2022-23 Fed hiking cycle; historically, market depth has displayed a non-linear, inverse relationship with volatility (**Figure 58**).

**Figure 57: Market depth has steadily improved from its nadir in 2023...**

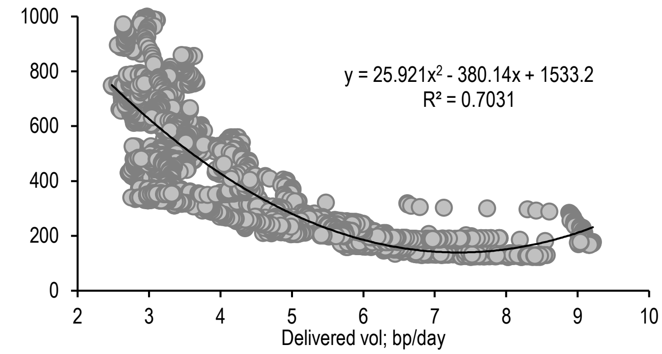
Duration-weighted Treasury market depth*; $mn 10-year Treasury equivalents




*Sum of the three bids and offers by queue position, averaged between 8:30-10:30am. This is the sume of 2-, 5-, 10-, and 30-year depth normalized by 10-year duration
Source: J.P. Morgan, BrokerTec

**Figure 58: ...as volatility has declined off of its highs associated with the 2022-23 Fed hiking cycle**

3-month moving average of duration-weighted Treasury market depth*, regressed on 3-month standard deviation of daily changes in US-GBI (bp/day); regression over the past decade; $mn 10-year equivalents

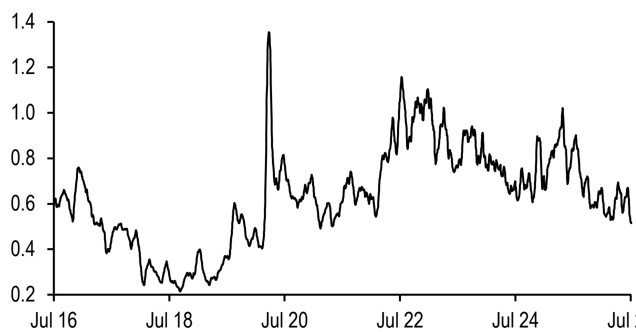



*See Figure 57 for more details
Source: J.P. Morgan, BrokerTec

Away from market depth, we observe evidence of improved liquidity and market functioning across other metrics we track. **Figure 59** displays the evolution of price impact, which measures the average move for a $100mn notional trade in 10-year Treasuries. This metric assesses liquidity through the lens of the ease of transaction, based on the empirical findings that trades of a standardized notional size have a larger impact on market pricing when there is an imbalance in bid/ask volumes on the CLOB (see *Drivers of price impact and the role of hidden liquidity*, 1/13/17). Unsurprisingly, price impact rose as volatility increased during the Fed's 2022-23 tightening campaign, and again in the wake of the spring 2025 "Liberation Day" announcement as policy uncertainty rose, but has resumed its downward trend as volatility has subsided.

**Figure 59: Price impact has declined, indicating a smaller footprint for each trade in the Treasury market...**

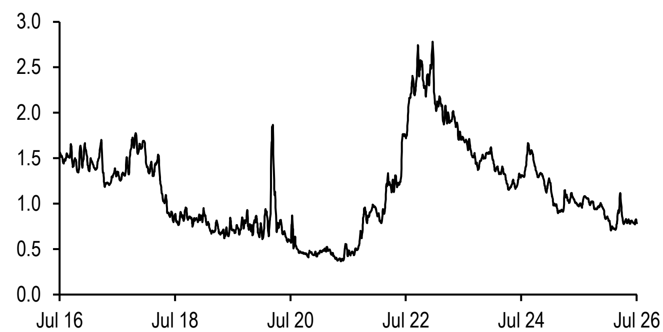
10-year Treasury price impact*; 1-month moving average; 32nds




*The average move in order book mid-price against a $100mn flow in traded notional. See *Drivers of price impact and the role of hidden liquidity*, 1/13/17 for more details
Source: J.P. Morgan, BrokerTec

**Figure 60: ...and dispersion relative to our par curve has declined, suggesting balance-sheet intensive off-the-runs are under little stress**

Root mean squared error of the J.P. Morgan par curve, one week moving average; bp




Source: J.P. Morgan

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In contrast to on-the-runs, where bid/offer spreads, trading volumes, and market depth are readily available, only limited data exist for off-the-run securities. While FINRA's TRACE data offer trading volumes for off-the-run Treasuries broken down by various maturity buckets, there is little insight into how individual off-the-runs trade. In light of this, we can instead look at indirect metrics: **Figure 60** shows the Root Mean Squared Error (RMSE) of the Treasury curve relative to our par-fitted curve aggregated over the past decade. Like previous metrics, dispersion rose during the Fed's recent massive tightening campaign, but has drifted lower in recent years alongside lower volatility and the aforementioned efforts to ease balance sheet constraints, indicating increased risk appetite and easier transaction costs in these more illiquid securities.

Putting the pieces together, we construct a broad index of market functioning in **Figure 61**. This index displays similar trends to those noted above, with functioning sitting considerably above its three-year average, especially at the long end of the curve. This is not surprising to the extent that volatility has remained lower at the long-end in recent months, we have observed increased participation of HFTs at the long end, and there have been improvements to Treasury's buyback program (see *Ain't nothin' gonna break my stride: Treasury market liquidity continues to improve*, 2/27/26).

**Figure 61: An array of liquidity metrics indicate that Treasury market functioning is considerably above medium-term average levels across the curve**

Current levels and 3-year z-scores for Treasury market depth*, price impact**, and Treasury curve RMSE***, by sector of the curve, with Treasury market functioning index†; units as indicated

| | Market depth ($mn) | | Price impact (32nds) | | RMSE (bp) | | Market functioning index |
|---|---|---|---|---|---|---|---|
| | Current | 3y z-score | Current | 3y z-score | Current | 3y z-score | |
| 2y | 240 | 3.1 | 0.25 | 0.2 | 1.1 | 1.5 | 1.6 |
| 5y | 193 | 2.7 | 0.19 | 1.7 | 0.9 | 0.3 | 1.6 |
| 10y | 339 | 4.0 | 0.60 | 0.8 | 1.9 | -0.3 | 1.5 |
| 30y | 40 | 3.0 | 0.50 | 1.0 | 0.7 | 1.9 | 2.0 |

*Market depth is the sum of the three bids and offers by queue position, averaged between 8:30 and 10:30 am daily
**Defined as passive limit orders that are reacting within 0.1 second to another event in the CLOB
***Root Mean Squared Error of the J.P. Morgan par fitted Treasury curve. We use Fed purchase buckets: 2y = 0-2y, 5y = 3-5y, 10y = 7-10y, 30y = 20-30y
† We multiply the price impact and RMSE z-scores by -1 so positive z-score means better Treasury market functioning
Source: J.P. Morgan, BrokerTec

Treasury market liquidity has also been bolstered through the efforts of the IAWG over the past five years. Ordinarily, during times of heightened uncertainty, demand for Treasuries increases and yields decline as investors flock to safe, liquid assets as a store of value. However, during the onset of the COVID-19 pandemic, yields actually rose as investors sold even the most liquid products to meet their demands for liquidity in what has been dubbed the 'Dash for Cash', requiring the Fed to bolster markets through an unprecedented pace of purchases. Shortly thereafter, the IAWG met to outline five workstreams focused on improving market stability, resilience, transparency, efficiency, and data quality. Many of the policies discussed in this primer are a direct result of this effort, including the introduction of the Treasury buyback program, the adoption of a central clearing mandate, and increased enhancements to TRACE data quality. When combined with efforts to reduce and streamline bank capital controls, policy makers have attuned to the growing risks to Treasuries' liquidity value. However, as the Treasury market continues to grow, more work will be needed to ensure the Treasury market remains the largest and deepest fixed income market in the world.

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